Exercise list — Topic 4: Moral hazard, credit rationing, asset substitution, debt overhang

Worked solutions

Statements are in `Listas/lista-topic4-moral-hazard-overhang.md` (and its PDF); each exercise is cited by number and printed page and is not restated. Exercises 16–19 are original exercises (no book exercise exists; written with the user's agreement, 2026-11-12), built on Prof. Ricca's Topic 4B slides 37–58; each opens with a one-paragraph data line so that it reads on its own, and the data of exercise 19 are hypothetical. Short answers: `lista-topic4-moral-hazard-overhang-solutions-short.pdf`. Editions are the local copies: Tirole, *The Theory of Corporate Finance* (2005); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001); Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Grinblatt–Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013).

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1. DIP financing and existing creditors — Grinblatt & Titman 16.14, p. 593

What is asked. A single conceptual question. Chapter 11 lets a bankrupt firm borrow new money that is paid *before* the old debt. On its face this hurts the old creditors, who move down the queue. Explain the mechanism by which they can nevertheless end up better off.

Approach. This is the lecture's *debt overhang* problem (Myers 1977). Write every claim as a state-by-state payoff, debt = $\min\{\text{assets}, F\}$ and equity = $\max\{\text{assets} - F, 0\}$, and compare the old creditors' payoff with and without the new project. Junior money cannot be raised when the old debt is under water, because the old creditors capture the project's value; seniority for the new money removes that obstacle. The old creditors lose priority but gain the project's cash flows, so the comparison is "smaller share of a larger pie" versus "larger share of a smaller pie".

Notation.

- s: state of the world at the payoff date; $V_s \geq 0$: value of the assets in place in state s.
- $F > 0$: face value of the existing debt, protected by a covenant that forbids issuing debt senior to it.
- $I > 0$: cost of a new project that must be carried out inside the firm; $X_s \geq 0$: its payoff in state s; risk-neutral investors, zero riskless rate (as in the book's and the lecture's examples).
- N: face value of the new debtor-in-possession (DIP) debt, senior to F; D, E: market values of the existing debt and of equity.

Step 1 — Why the project is not funded outside bankruptcy. We first show that, without a seniority exception, a good project is passed up; this is the status quo the old creditors compare against. Without DIP financing, new money can only be junior to F. In state s a junior claimant receives at most what is left after the senior debt,

$$\text{junior payoff}_s \leq \max\{V_s + X_s - F,\, 0\}. \tag{1}$$

In the book's Lily Pharmaceuticals example (pp. 564–566), in the unfavourable state the assets in place are worth nothing without further development ($V_s = 0$), development costs $I = 100$ and yields $X_s = 150$, and the senior obligation is $F = 110$. Substituting into (1),

$$\begin{aligned} \text{junior payoff} &\leq \max\{(0) + (150) - (110),\, 0\} \\ &= \max\{40,\, 0\} \\ &= 40 < 100 = I, \end{aligned}$$

so no one lends: junior lenders would pay 100 for a claim worth at most 40. Yet the project is worth doing,

$$\begin{aligned} NPV &= X_s - I \\ &= (150) - (100) \\ &= 50 > 0, \end{aligned}$$

so it is passed up, the firm defaults, and the existing lenders receive $\min\{V_s, F\} = \min\{0, 110\} = 0$ in that state.

Step 2 — DIP debt makes it fundable. We now let the new money jump the queue and recompute the old creditors' payoff. In Chapter 11 the firm may issue debt senior to F. A DIP loan

of $I = 100$ at the zero riskless rate has face $N = 100$ and is riskless because $150 \geq 100$. The existing lenders now receive

$$\begin{aligned} \text{existing debt payoff} &= \min\{V_s + X_s - N,\, F\} \\ &= \min\{(0) + (150) - (100),\, (110)\} \\ &= \min\{50,\, 110\} \\ &= 50, \end{aligned} \tag{2}$$

instead of 0: they gain $50 - 0 = 50$, the whole NPV, even though they lost their priority.

Step 3 — The general argument. The example is not special; we show the old creditors never lose when the DIP money funds a project that pays back at least its cost. With a riskless DIP loan ($N = I \leq X_s$ in every state), the existing debt's payoff with the project is $\min\{V_s + X_s - I,\, F\}$, and without it $\min\{V_s,\, F\}$. Since $X_s - I \geq 0$, the first argument of the minimum is larger with the project, $V_s + X_s - I \geq V_s$, and $\min\{\cdot, F\}$ is nondecreasing in its first argument, so

$$\min\{V_s + X_s - I,\, F\} \geq \min\{V_s,\, F\} \qquad \text{for every } s, \tag{3}$$

with strict inequality in every state where the old debt was impaired ($V_s < F$). Seniority is diluted, but the asset pool the old creditors are paid from grows by more than the new claim.

Step 4 — The lecture's numbers. Topic 4B (slides 22–31): assets in place worth 500 (boom) or 200 (slump), each with probability 0.5; $F = 300$; a project costs $I = 60$ and adds 100 in both states. We value both claims without the project, then with it. Without the project,

$$\begin{aligned} D &= (0.5)\min\{500, 300\} + (0.5)\min\{200, 300\} \\ &= (0.5)(300) + (0.5)(200) = 150 + 100 = 250, \\ E &= (0.5)\max\{500 - 300, 0\} + (0.5)\max\{200 - 300, 0\} \\ &= (0.5)(200) + (0.5)(0) = 100. \end{aligned}$$

Financing the 60 with senior DIP debt ($N = 60$, riskless since the slump assets $200 + 100 = 300 \geq 60$), the assets become $500 + 100 = 600$ (boom) and $200 + 100 = 300$ (slump), and the old debt receives, by (2),

$$\begin{aligned} \text{boom:}\ \ &\min\{(600) - (60),\, (300)\} = \min\{540, 300\} = 300, \\ \text{slump:}\ \ &\min\{(300) - (60),\, (300)\} = \min\{240, 300\} = 240, \\ D &= (0.5)(300) + (0.5)(240) \\ &= 150 + 120 = 270, \end{aligned} \tag{4}$$

and equity receives $\max\{600 - 60 - 300, 0\} = 240$ (boom) and $\max\{300 - 60 - 300, 0\} = 0$ (slump), so $E = (0.5)(240) + (0.5)(0) = 120$. Both claims rise by 20 ($270 - 250$ and $120 - 100$), and $20 + 20 = 40 = 100 - 60$ is the project's NPV.

Answer (16.14).

project cannot be financed (debt overhang) and they are paid out of the assets in place only. With DIP financing the project is done; the DIP lender is repaid first, but the project adds more to the pool than the DIP claim takes out ($X_s \geq I$), so the old creditors' payoff rises in every impaired state (\$0 to \$50 in the Lily example; \$250 to \$270 in the lecture's example).

Step 5 — The limit. (3) needs $X_s \geq I$: a good, safe project. The same provision lets a distressed firm borrow to keep operating when it would be worth more liquidated; then $X_s < I$ in some state and the old creditors lose (Result 16.11).

Intuition. Debt overhang means the old creditors capture part of the value of new investment while the new money bears its full cost, so junior money cannot be raised. Letting new money jump the queue removes the overhang. The old creditors, who would otherwise get little or nothing,

take a smaller slice of a larger pie. A covenant that blocks this hurts them, and a diffuse group of bondholders cannot easily agree to lend the money themselves (free-rider problem). Chapter 11 unlocks senior financing only in distress, which keeps most of the protection the covenant gives in normal times.

Common mistakes.

- Comparing the old creditors' payoff with the project under DIP financing against their payoff with the project under *junior* financing. The relevant status quo is no project at all, because junior money cannot be raised.
- Crediting the old creditors with the project's gross payoff X_s and forgetting that the DIP lender is repaid $N = I$ first; their gain is bounded by $X_s - I$, the project's NPV.
- Stating that DIP financing always helps creditors. It helps only when the funded project pays back at least its cost ($X_s \geq I$); financing a value-destroying continuation hurts them.

Ref: Grinblatt & Titman Ch. 16, § 16.2 "Underinvestment as a Consequence of Debt Overhang" (Lily example), pp. 564–566; § 16.3 "How Chapter 11 Bankruptcy Mitigates Debt Holder–Equity Holder Incentive Problems", Result 16.11, p. 580. Lecture Topic 4B, slides 22–31.

2. Asset substitution: project A or B — Grinblatt & Titman 16.1, p. 591

What is asked. (a) The net present value of each project for the firm as a whole. (b) Which project the shareholders, who control the firm, would pick, and the reason for their choice.

Approach. With risk-neutral investors and a zero riskless rate, a claim's value is its expected payoff. Debt and equity split each state's payoff: debt gets $\min\{X, F\}$, equity gets $\max\{X - F, 0\}$ (limited liability). Because the cash on hand exactly equals the face value, equity is an at-the-money call option on the project payoff, so it values variance. This is the lecture's *asset substitution* (risk-shifting) problem: comparing firm value (a) with equity value (b) shows the gap between what is efficient and what shareholders choose.

Notation.

- $j \in \{A, B\}$: project; $X_{j,s}$: its payoff in state $s \in \{\text{fav}, \text{unf}\}$, each with probability $\pi = 0.5$.
- $C = 100$: cash on hand, all spent on the project; $F = 100$: debt due next period; $r_f = 0$; risk-neutral investors.
- $V_j = \mathbb{E}[X_j]$: firm value under j; E_j, D_j: values of equity and debt under j.

Step 1 — (a) NPVs. We value each project's payoff and subtract the \$100 it costs. With $r_f = 0$ and risk neutrality the present value is the expected payoff:

$$\begin{aligned} V_A &= \pi X_{A,\text{fav}} + (1-\pi) X_{A,\text{unf}} \qquad (5)\\ &= (0.5)(120) + (0.5)(60) \\ &= 60 + 30 = 90, \\ NPV_A &= V_A - C \qquad (6)\\ &= (90) - (100) \\ &= -10, \\ V_B &= \pi X_{B,\text{fav}} + (1-\pi) X_{B,\text{unf}} \qquad (7)\\ &= (0.5)(101) + (0.5)(101) \\ &= 50.5 + 50.5 = 101, \\ NPV_B &= V_B - C \\ &= (101) - (100) \\ &= 1. \end{aligned}$$

Answer (a). $NPV_A = -\$10$ million, $NPV_B = +\$1$ million: only B creates value.

Step 2 — (b) Payoffs to each claim, state by state. The shareholders care about their own claim, not about firm value, so we split each project's payoff between the claims. Equity receives $\max\{X - F, 0\}$ and debt $\min\{X, F\}$:

	Equity		Debt	
	Favourable	Unfavourable	Favourable	Unfavourable
A	$120 - 100 = 20$	0	100	60
B	$101 - 100 = 1$	$101 - 100 = 1$	100	100

(In A's unfavourable state $\max\{60 - 100, 0\} = 0$ and $\min\{60, 100\} = 60$.) Taking expectations,

$$E_A = (0.5)(20) + (0.5)(0) = 10 + 0 = 10, \qquad D_A = (0.5)(100) + (0.5)(60) = 50 + 30 = 80, \tag{8}$$

$$E_B = (0.5)(1) + (0.5)(1) = 0.5 + 0.5 = 1, \qquad D_B = (0.5)(100) + (0.5)(100) = 100. \tag{9}$$

Check: $E_A + D_A = 10 + 80 = 90 = V_A$ and $E_B + D_B = 1 + 100 = 101 = V_B$.

Step 3 — (b) The choice. We compare the shareholders' payoffs and decompose the difference into a value effect and a transfer. Equity holders, who control the firm, want project A: $E_A = 10 > E_B = 1$, although A destroys value. Writing $E_j = V_j - D_j$ and using (8)–(9),

$$\begin{aligned} E_A - E_B &= (V_A - V_B) - (D_A - D_B) \\ &= (90 - 101) - (80 - 100) \\ &= (-11) - (-20) \\ &= 9 : \end{aligned} \tag{10}$$

switching to A lowers firm value by 11 but moves 20 from the creditors to the shareholders.

Answer (b). The shareholders want project A ($E_A = \$10$ million $> E_B = \$1$ million), because limited liability lets them keep A's upside while the creditors bear its downside: the \$20 million transferred from debt exceeds the \$11 million of value destroyed.

Intuition. With \$100 due and \$100 in cash, the equity is an at-the-money call on the firm's assets. The safe project leaves it almost worthless (1); the risky one gives it the whole upside (20) while the creditors absorb the downside (60 instead of 100). This is the lecture's asset-substitution (risk-shifting) problem: limited liability makes levered equity prefer variance, even at negative NPV.

Common mistakes.

- Writing equity's payoff as $X - F$ in every state. In A's unfavourable state that gives -40; limited liability caps the loss at 0, and that cap is exactly what makes A attractive.
- Answering (b) with the NPVs of (a). Firm value and equity value rank the projects in opposite orders; (b) asks about the shareholders' claim.
- Forgetting that the \$100 of cash is spent on the project: the NPV subtracts it, but the payoff split in (b) uses only the project payoff and the \$100 face value.

Ref: Grinblatt & Titman Ch. 16, § 16.2 "The Asset Substitution Problem", pp. 569–572 (Results 16.7–16.8). Lecture Topic 4B, slides 2–8.

3. Debt overhang on vacant land — Berk 16.25, pp. 494–495

What is asked. (a) Today's values of equity and debt if the land stays vacant. (b) The NPV of developing the land, for the firm as a whole. (c) Today's values of equity and debt if the shareholders put in \$20 million and the land is developed. (d) Whether the shareholders gain from putting in that money, given (c).

Approach. All cash flows are riskless, so every claim is its payoff in one year discounted at $r_f = 10\%$; debt gets $\min\{\text{land value}, 15\}$ and equity the rest. This is the lecture's *debt overhang* (underinvestment) problem: the debt is under water, so part of the value a new project creates goes to the creditors, who pay none of its cost. Comparing the NPV (b) with the shareholders' gain (d) shows the wedge.

Notation.

- $F = 15$: face value of the debt, due in one year; $r_f = 0.10$; all cash flows riskless; no taxes.
- $L_0 = 10$: value of the vacant land in one year; $L_1 = 35$: value of the developed land in one year; $K = 20$: development cost paid today.
- E, D: values today of equity and debt.

Step 1 — (a) Land left vacant. We split the land's value in one year between the claims and discount. In one year the firm is worth $10 < 15$: the creditors take 10 and the shareholders get 0. Discounting at r_f,

$$\begin{aligned} D &= \frac{\min\{L_0, F\}}{1 + r_f} \\ &= \frac{\min\{10, 15\}}{1 + 0.10} \\ &= \frac{(10)}{(1.10)} \\ &= 9.0909, \\ E &= \frac{\max\{L_0 - F, 0\}}{1 + r_f} = \frac{\max\{10 - 15, 0\}}{1.10} = \frac{(0)}{(1.10)} = 0. \end{aligned} \tag{11}$$

Answer (a). Equity is worth \$0; debt is worth \$9.09 million.

Step 2 — (b) NPV of developing. The NPV counts only the cash flows that change with the decision. Developing replaces land worth 10 by land worth 35, so the incremental cash flow in one year is $35 - 10 = 25$, against 20 paid today:

$$\begin{aligned} NPV &= -K + \frac{L_1 - L_0}{1 + r_f} \\ &= -(20) + \frac{(35) - (10)}{(1.10)} \\ &= -20 + \frac{25}{1.10} \\ &= -20 + 22.7273 \\ &= 2.7273. \end{aligned} \tag{12}$$

(Ignoring the vacant-land value forgone would give $-20 + 35/1.10 = 11.8182$, the NPV measured against a firm with no land at all; the decision-relevant figure is (12).)

Answer (b). $NPV = \$2.73$ million > 0: developing creates value.

Step 3 — (c) Values if developed with \$20 of new equity. We repeat Step 1 with the developed land. The \$20 is spent on development, so it is not an asset in one year. In one year the firm is worth $35 \geq 15$: the debt is paid in full, $\min\{35, 15\} = 15$, and equity keeps $\max\{35 - 15, 0\} = 20$.

Hence

$$D = \frac{(15)}{(1.10)} \tag{13}$$
$$= 13.6364,$$
$$E = \frac{(35) - (15)}{(1.10)} \tag{14}$$
$$= \frac{20}{1.10}$$
$$= 18.1818.$$

Answer (c). Equity is worth \$18.18 million and debt \$13.64 million today.

Step 4 — (d) Will the shareholders put up \$20?. The shareholders compare what they hold after contributing with what they hold if they do nothing. Their gain is the equity value (14), minus the equity value without development (Step 1), minus the contribution:

$$\begin{aligned}\Delta E &= E_{\text{dev}} - E_{\text{vacant}} - K \\ &= (18.1818) - (0) - (20) \\ &= -1.8182 < 0, \\ \Delta D &= D_{\text{dev}} - D_{\text{vacant}} \\ &= (13.6364) - (9.0909) \\ &= 4.5455.\end{aligned}$$

No: they would lose \$1.82 million. The sum $-1.8182 + 4.5455 = 2.7273$ is the NPV (12); the creditors capture more than all of it.

Answer (d). No. The shareholders would pay \$20 million for an equity stake worth \$18.18 million (a loss of \$1.82 million), while the debt gains \$4.55 million; the positive-NPV project is passed up.

Intuition. The debt is under water ($10 < 15$), so the first \$5 million of any value the project creates goes to the creditors, who pay none of its cost. The shareholders bear the \$20 million and receive only the value above the \$15 million face. This is Myers' debt overhang: leverage acts as a tax on new investment, and a positive-NPV project is passed up. Writing the debt down (renegotiation, as in the lecture) or senior new financing would fix it.

Common mistakes.

- Computing the NPV as $-20 + 35/1.10 = 11.82$: the vacant land worth \$10 million in one year is given up, so it is part of the project's cost.
- Answering (d) with $E_{\text{dev}} = 18.18 > 0$, "so yes". The shareholders must compare the stake with the \$20 million they put in.
- Valuing the debt in (a) at the face value 15/1.10; the creditors can get at most what the land is worth, $\min\{10, 15\}$.

Ref: Berk Ch. 16, § 16.6 "Agency Costs and Information", "Equity-Debt Holder Conflicts" (under-investment), pp. 483–484. Lecture Topic 4B, slides 19–30.

4. Short-term debt and incentive problems — Grinblatt & Titman 16.11, p. 593

What is asked. A single conceptual question: why the conflicts between shareholders and creditors (asset substitution, debt overhang, under-maintenance) do less damage when the firm finances itself with debt that matures soon and must be rolled over, rather than with long-term debt.

Approach. The conflicts exist because equity chooses the firm's strategy *after* the terms of the debt are fixed, so it can change the value of the creditors' claim (the lecture's asset substitution and debt overhang). We write equity as firm value minus debt value and ask what happens to the debt term when the debt is repriced after the choice is made: if new lenders price the debt knowing the strategy, the debt term no longer depends on the strategy and equity internalizes the full value consequences.

Notation.

- j: investment strategy (project choice, risk level, maintenance), chosen by the equity holders after the debt is in place; $X_j \geq 0$: resulting terminal firm value, $V_j = \mathbb{E}[X_j]$ (risk neutrality, $r_f = 0$).
- F: face value of the debt; L: amount the firm owes when its short-term debt is rolled over; D_j, E_j: values of debt and equity under j.

Step 1 — Where the conflict comes from. We locate the conflict in the debt term of equity's value. With long-term debt, F is fixed before j is chosen, and since $\max\{X - F, 0\} = X - \min\{X, F\}$ in every state,

$$\begin{aligned} E_j &= \mathbb{E}[\max\{X_j - F, 0\}] \\ &= V_j - D_j, \qquad D_j = \mathbb{E}[\min\{X_j, F\}]. \end{aligned} \tag{15}$$

Equity can gain by picking a j that lowers D_j (more risk, less investment, payouts) even when V_j falls: comparing two strategies, $E_j - E_k = (V_j - V_k) - (D_j - D_k)$, and the second term is a transfer from creditors whenever $D_j < D_k$.

Step 2 — Debt that is repriced when the choice is made. We now let the debt be repriced after the choice and show the transfer term disappears. If the debt matures before or at the decision and must be rolled over, the new lenders see j and set the new face F_j so that their claim is worth what they lend, $\mathbb{E}[\min\{X_j, F_j\}] = L$. Substituting into (15),

$$E_j = V_j - (L), \tag{16}$$

and L does not depend on j: equity bears the full consequences of its choice and picks the value-maximizing strategy. Short-term debt is worth close to its face whatever the firm does, so there is little value to expropriate.

Step 3 — Illustration (Grinblatt & Titman 16.1). Take the firm of G&T Exercise 16.1: \$100 of debt, projects A (pays 120 or 60) and B (pays 101 in both states), each state with probability 0.5, $r_f = 0$. With long-term debt, equity is worth $0.5(20) + 0.5(0) = 10$ under A and $0.5(1) + 0.5(1) = 1$ under B, so it picks the value-destroying A. Suppose instead the \$100 had to be refinanced after the project is chosen. Firm values are $V_A = 0.5(120) + 0.5(60) = 90$ and $V_B = 0.5(101) + 0.5(101) = 101$; by (16) with $L = 100$,

$$E_A = (90) - (100) = -10, \qquad E_B = (101) - (100) = 1.$$

Under A the firm cannot raise \$100 (no claim on payoffs worth 90 is worth 100), so the equity holders choose B, the positive-NPV project, instead of A.

Step 4 — Debt overhang. Myers (1977): if the existing debt matures before the firm must raise money for a new project, the debt is repriced together with the project decision. The project's benefit to creditors (a safer firm) then shows up as a lower borrowing rate that accrues to equity, and positive-NPV projects are taken. In Unitron's case, the riskless 12% project would lower the firm's 13% borrowing rate and become attractive.

Step 5 — Asset substitution. The lending rate is renegotiated after the risk choice is made and seen, so a risky choice raises the firm's own cost of funds instead of transferring wealth from existing lenders. The threat of not being rolled over also disciplines the choice (lecture Topic 4B, slide 16).

Step 6 — The cost. Rolling over exposes the firm to refinancing and interest-rate risk: an unexpected rise in rates can push a highly levered firm into distress (hence the advice to pair short-term debt with interest-rate hedges).

Answer (16.11). Short-term debt is repriced (rolled over) after the shareholders' investment and risk choices are made and observed, so its terms reflect those choices: equity bears the full value consequences, $E_j = V_j - L$, and cannot shift wealth from creditors through risk-taking or under-investment. Short-term debt is also close to riskless, so there is little value to expropriate. The price is rollover and interest-rate risk.

Intuition. The conflicts exist because equity can change the value of a claim whose terms are already fixed. The shorter the maturity, the sooner those terms are reset at prices that reflect what equity has done, and the less wealth equity can take from creditors. The incentive problem shrinks, and the firm takes on rollover risk in exchange.

Common mistakes.

- Saying short-term debt helps "because it is safer". The point is timing: the debt is priced after the decision, so the decision's effect on the creditors comes back to equity through the new rate.
- Confusing maturity with seniority. Seniority decides who is paid first in default; maturity decides when the terms are reset.
- Omitting the cost. Short maturities expose the firm to refinancing and interest-rate risk, so they are not a free cure.

Ref: Grinblatt & Titman Ch. 16, § 16.4 "The Use of Short-Term versus Long-Term Debt", pp. 584–585 (Result 16.12). Lecture Topic 4B, slide 16.

5. Random financing — Tirole 3.1, p. 144

What is asked. (i) Write the condition under which investors are willing to sign a contract in which the borrower contributes $\hat{A}$ of her cash, the project then goes ahead only with probability x, and she gets R_b on success. (ii) Show that a borrower with too little cash for certain financing ($A < \bar{A}$) should put *all* her cash in, and describe how the financing probability depends on A.

Approach. This is the Holmström–Tirole fixed-investment model of the lecture (Topic 4A). The two constraints are the borrower's incentive constraint (IC_b), which caps the *pledgeable income* at $\mathcal{P} = p_H(R - B/\Delta p)$, and the investors' breakeven constraint (IR_l). Below $\bar{A}$ the project cannot be financed for sure; a lottery over financing scales the investors' outlay and the pledgeable income by x but not the borrower's contribution, which convexifies the indivisible project. We write (IR_l) for the lottery, show (IC_b) must hold, and maximise the borrower's utility over $(x, \hat{A})$.

Notation.

- $I > 0$: fixed investment; $A \in (0, \bar{A})$: borrower's cash; $\hat{A} \in [0, A]$: the part she contributes; $x \in [0, 1]$: probability that the project is financed.
- $R > 0$: income on success (0 on failure); $R_b \geq 0$: borrower's reward on success; $R_l = R - R_b$: investors' claim on success.
- $p_H > p_L$: success probabilities when she behaves / misbehaves; $\Delta p = p_H - p_L$; $B > 0$: private benefit from misbehaving.
- $\mathcal{P} = p_H(R - B/\Delta p)$: pledgeable income; $\bar{A} = I - \mathcal{P}$: minimum cash for financing with probability 1 (the statement's definition); U_b: borrower's net utility. Rate of interest 0, everybody risk neutral.

Assumption. The borrower hands over $\hat{A}$ when the contract is signed, whether or not the lottery then selects the project (this is the reading in Tirole's answer). If $\hat{A}$ were paid only when the project

goes ahead, the expected outlay would be $x(I-\hat{A})$ and the breakeven condition $x(I-\hat{A}) \le x\mathcal{P}$ would require $\hat{A} \ge \bar{A} > A$: randomizing would achieve nothing.

Step 1 — (i) Investors' breakeven condition. Investors receive $\hat{A}$ for sure, pay I with probability x, and, if the borrower behaves, collect R_l with probability p_H when the project runs:

$$\underbrace{\hat{A}}_{\text{received}} + \underbrace{x\,p_H(R-R_b)}_{\text{expected repayment}} \ge \underbrace{xI}_{\text{expected outlay}}$$

$$x\,p_H(R-R_b) \ge xI - \hat{A}. \tag{17}$$

Answer (i). $x\,p_H(R-R_b) \ge xI - \hat{A}$ (with $R_b \ge B/\Delta p$ so that she behaves).

Step 2 — The contract must induce good behaviour. We check that a contract under which she shirks is never worth signing, so (IC_b) can be imposed. Suppose $(\Delta p)R_b < B$, so she shirks. Investors break even only if $xp_L(R-R_b) \ge xI - \hat{A}$, i.e. $-\hat{A} \le xp_L(R-R_b) - xI$. Her net utility is then

$$\begin{aligned} x(p_L R_b + B) - \hat{A} &\le x(p_L R_b + B) + \big(xp_L(R-R_b) - xI\big) \\ &= xp_L R_b + xB + xp_L R - xp_L R_b - xI \\ &= x(p_L R + B - I) < 0 \end{aligned} \tag{18}$$

by (3.2), worse than not signing. So the contract satisfies (IC_b), $(\Delta p)R_b \ge B$, i.e. $R_b \ge B/\Delta p$.

Step 3 — Breakeven in terms of $\bar{A}$. We combine the two constraints into one feasibility condition on $(x, \hat{A})$. (IC_b) gives $R - R_b \le R - B/\Delta p$, so the left side of (17) is at most $x\,p_H(R - B/\Delta p) = x\mathcal{P}$. A contract exists for $(x, \hat{A})$ only if

$$\begin{aligned} xI - \hat{A} &\le x\,p_H\left(R - \frac{B}{\Delta p}\right) \\ xI - \hat{A} &\le x(\mathcal{P}) \\ x(I-\mathcal{P}) &\le \hat{A} \\ x(\bar{A}) &\le \hat{A} \\ x &\le \frac{\hat{A}}{\bar{A}}, \end{aligned} \tag{19}$$

substituting $I - \mathcal{P} = \bar{A}$ from the statement's definition (third to fourth line) and dividing by $\bar{A} > 0$ (last line). Conversely, if (19) holds, $R_b = B/\Delta p$ satisfies both constraints, so (19) is necessary and sufficient.

Step 4 — (ii) The borrower's objective. We write her utility and use the binding breakeven constraint to express it in terms of x only. She keeps $A - \hat{A}$ and, with probability x, earns R_b with probability p_H; net of the A she would consume anyway,

$$\begin{aligned} U_b &= x\,p_H R_b + (A - \hat{A}) - A \\ &= x\,p_H R_b - \hat{A}. \end{aligned} \tag{20}$$

Competitive investors make (17) bind; expanding and solving for $xp_H R_b$,

$$\begin{aligned} xp_H R - xp_H R_b &= xI - \hat{A} \\ xp_H R_b &= xp_H R - xI + \hat{A}. \end{aligned}$$

Substituting into (20),

$$\begin{aligned} U_b &= (xp_H R - xI + \hat{A}) - \hat{A} \\ &= x\,(p_H R - I). \end{aligned} \tag{21}$$

Step 5 — (ii) Optimal $\hat{A}$ and x. We maximise (21) over the feasible set (19). Differentiating,

$$\frac{\partial U_b}{\partial x} = p_H R - I > 0, \qquad \frac{\partial U_b}{\partial \hat{A}} = 0,$$

so U_b rises with x and $\hat{A}$ matters only through feasibility. By (19), the largest feasible x is $\hat{A}/\bar{A}$ (below 1 because $\hat{A} \le A < \bar{A}$), which increases with $\hat{A}$: substituting it, $U_b = (\hat{A}/\bar{A})(p_H R - I)$, increasing in $\hat{A}$ on $[0, A]$. So she contributes everything:

$$\hat{A}^* = A,$$
$$x^* = \frac{A}{\bar{A}}, \tag{22}$$
$$U_b^* = \frac{A}{\bar{A}}\,(p_H R - I). \tag{23}$$

At x^* the binding (17), divided by $x^* = A/\bar{A}$, gives $p_H(R - R_b) = I - A/x^* = I - \bar{A} = \mathcal{P}$, so $R - R_b = R - B/\Delta p$ and $R_b = B/\Delta p$: (IC$_b$) binds.

Step 6 — (ii) How x^* varies with A. $x^* = A/\bar{A}$ rises linearly from 0 at $A = 0$ to 1 at $A = \bar{A}$, and stays at 1 above $\bar{A}$. Utility (23) therefore rises smoothly to the full NPV instead of jumping at $\bar{A}$ as in (3.6).

Numerical illustration. $I = 100$, $R = 150$, $p_H = 0.8$, $p_L = 0.4$, $B = 30$, $A = 10$: (3.2) holds ($0.4(150) - 100 + 30 = -10 < 0$); $B/\Delta p = 75$; $\mathcal{P} = 0.8(150 - 75) = 60$; $\bar{A} = 100 - 60 = 40$; NPV $= 0.8(150) - 100 = 20$. Then $x^* = 10/40 = 0.25$ and $U_b^* = 0.25(20) = 5$; investors pay $0.25(100) - 10 = 15$ in expectation and receive $0.25(60) = 15$.

Answer (ii). $\hat{A}^* = A$; the financing probability is $x^* = A/\bar{A}$, rising linearly from 0 at $A = 0$ to 1 at $A = \bar{A}$ (and $x^* = 1$ for $A \ge \bar{A}$); her utility is $x^*(p_H R - I)$.

Intuition. The obstacle is not the size of the NPV but the pledgeable income per unit of investment. A poor borrower can still buy a *fraction* of the deal: each unit of her cash covers the investors' shortfall on $1/\bar{A}$ units of probability of financing. Since she captures the whole NPV whenever the project runs, she wants as high a probability as her cash can buy and puts all of it in. Lotteries convexify the indivisible project.

Common mistakes.

- Writing the breakeven as $x p_H(R - R_b) \ge x(I - \hat{A})$, i.e. taking her contribution only when the project goes ahead. Then x cancels and randomisation achieves nothing.
- Using R instead of $R - B/\Delta p$ as the investors' maximal claim: without (IC$_b$) she shirks, and by (3.2) no one breaks even.
- Claiming she keeps some cash "for safety". Cash kept aside earns nothing extra, while each unit put in raises x by $1/\bar{A}$ and her utility by $(p_H R - I)/\bar{A} > 0$.

Book answer. Matches the book's answer (p. 571): breakeven $xI - A \le x p_H(R - R_b)$, equivalently $x\bar{A} \le A$; $U_b = x(p_H R - I)$, so $x^* = A/\bar{A}$, rising from 0 to 1 as A goes from 0 to $\bar{A}$.

Ref: Tirole Ch. 3, § 3.2.1–3.2.2 "The Fixed-Investment Model", "The Lenders' Credit Analysis", pp. 115–117 (footnote 9, p. 116). Lecture Topic 4A, slides 26–34.

6. Zymase: leverage and strategy choice — Berk 16.26, p. 495

What is asked. (a) Which strategy gives the largest expected payoff to the firm as a whole. (b) and (c) Which strategy gives the shareholders the largest expected payoff when \$40 million, respectively \$110 million, of debt is due at the payoff date. (d) How much expected firm value is lost, in each case, when management follows the shareholders' preference instead of the value-maximising strategy (the expected agency cost of debt).

Approach. Split each strategy's payoff state by state into debt, $\min\{X, F\}$, and equity, $\max\{X - F, 0\}$, and take expectations. Equity is a call option on the firm, so as F rises it favours strategies whose payoff lies above F with some probability, even if their mean is lower: the lecture's *asset substitution* (excessive risk-taking). The agency cost is the drop in expected *firm* payoff caused by that choice, which is the loss the shareholders bear ex ante when creditors price the debt.

Notation.

- $j \in \{A, B, C\}$: research strategy; X_j: its after-tax payoff (\$ million), with the probabilities in the table.
- $F \in \{40, 110\}$: face value of the debt, due when the payoff arrives; equity gets $\max\{X_j - F, 0\}$, debt $\min\{X_j, F\}$.
- Agency cost: expected firm payoff of the value-maximizing strategy minus that of the strategy equity chooses.

Assumption. Risk is diversifiable, so every claim is discounted at the riskless rate, which is not given. All comparisons are in expected payoffs at the payoff date; present values are these divided by the same factor $1 + r_f$, so the rankings do not change.

Step 1 — (a) Expected firm payoffs. We weight each payoff by its probability:

$$\mathbb{E}[X_A] = (1.0)(75) = 75, \tag{24}$$
$$\mathbb{E}[X_B] = (0.5)(140) + (0.5)(0) = 70 + 0 = 70, \tag{25}$$
$$\mathbb{E}[X_C] = (0.1)(300) + (0.9)(40) = 30 + 36 = 66. \tag{26}$$

Strategy A has the highest expected payoff (and the lowest risk).

Answer (a). Strategy A, with an expected payoff of \$75 million (B: \$70 million; C: \$66 million).

Step 2 — (b) Debt of \$40 million. We split each state's payoff with $F = 40$: equity $\max\{X - 40, 0\}$ (e.g. $140 - 40 = 100$; $\max\{40 - 40, 0\} = 0$), debt $\min\{X, 40\}$:

Strategy, state (prob.)	Firm	Equity $\max\{X - 40, 0\}$	Debt $\min\{X, 40\}$
A (1.0)	75	35	40
B, high (0.5)	140	100	40
B, low (0.5)	0	0	0
C, high (0.1)	300	260	40
C, low (0.9)	40	0	40

$$\begin{aligned} \mathbb{E}[E_A] &= (1.0)(35) = 35, \\ \mathbb{E}[E_B] &= (0.5)(100) + (0.5)(0) = 50 + 0 = 50, \\ \mathbb{E}[E_C] &= (0.1)(260) + (0.9)(0) = 26 + 0 = 26. \end{aligned} \tag{27}$$

Equity prefers B. Debt then expects $(0.5)(40) + (0.5)(0) = 20$, against 40 under A or C.

Answer (b). Strategy B (expected equity payoff \$50 million, against \$35 million for A and \$26 million for C).

Step 3 — (c) Debt of \$110 million. Same split with $F = 110$: equity $\max\{X - 110, 0\}$ (e.g. $\max\{75 - 110, 0\} = 0$, $140 - 110 = 30$, $300 - 110 = 190$), debt $\min\{X, 110\}$:

Strategy, state (prob.)	Firm	Equity $\max\{X - 110, 0\}$	Debt $\min\{X, 110\}$
A (1.0)	75	0	75
B, high (0.5)	140	30	110
B, low (0.5)	0	0	0
C, high (0.1)	300	190	110
C, low (0.9)	40	0	40

$$\begin{aligned}\mathbb{E}[E_A] &= (1.0)(0) = 0,\\ \mathbb{E}[E_B] &= (0.5)(30) + (0.5)(0) = 15 + 0 = 15,\\ \mathbb{E}[E_C] &= (0.1)(190) + (0.9)(0) = 19 + 0 = 19.\end{aligned} \tag{28}$$

Equity prefers C. Debt then expects $(0.1)(110) + (0.9)(40) = 11 + 36 = 47$, against 75 under A and $(0.5)(110) + (0.5)(0) = 55$ under B.

Answer (c). Strategy C (expected equity payoff \$19 million, against \$15 million for B and \$0 for A).

Step 4 — (d) Expected agency costs. The agency cost is the expected firm payoff of the efficient strategy minus that of the strategy equity picks. The value-maximizing strategy is A (24). With $F = 40$ equity picks B (25); with $F = 110$ it picks C (26):

$$\text{cost}(40) = \mathbb{E}[X_A] - \mathbb{E}[X_B] = (75) - (70) = 5, \tag{29}$$

$$\text{cost}(110) = \mathbb{E}[X_A] - \mathbb{E}[X_C] = (75) - (66) = 9. \tag{30}$$

The expected agency cost is \$5 million with \$40 million of debt and \$9 million with \$110 million (in payoff-date dollars; divide by $1 + r_f$ for today's value).

Answer (d). \$5 million with \$40 million of debt; \$9 million with \$110 million of debt (expected payoff-date values).

Intuition. With \$40 million due, A leaves equity a safe \$35 million, but B's 50% chance of \$100 million is worth more to equity because the creditors bear the zero-payoff state. With \$110 million due, A leaves equity nothing and only the long shot C pays off above the face value often enough. More debt pushes equity towards riskier, lower-value strategies, and the agency cost grows with leverage. Creditors who anticipate the choice price it in, so the shareholders bear the cost ex ante.

Common mistakes.

- Measuring the agency cost as the creditors' loss (e.g. $40 - 20 = 20$ with $F = 40$). That is a transfer; the agency cost is the loss in *total* expected payoff, $75 - 70 = 5$.
- Giving equity $X - F$ in states where $X < F$ (C's low state with $F = 110$ gives -70, not 0), which makes the risky strategies look worse to equity than they are.
- Comparing equity's payoffs across debt levels instead of across strategies at a given debt level; each of (b) and (c) is a choice among A, B, C with F fixed.

Ref: Berk Ch. 16, § 16.6 "Agency Costs and Information", "Equity-Debt Holder Conflicts" (excessive risk-taking), pp. 483–484. Lecture Topic 4B, slides 2–13.

7. Ajax Manufacturing: debt overhang — Grinblatt & Titman 16.12–16.13, p. 593

What is asked. (16.12) The NPV of the widget project, and whether Ajax can raise the \$75 million when the existing \$70 million senior bond forbids new debt (so only new equity is possible). (16.13) Whether Ajax can raise the \$75 million in equity if it uses the riskier, capital-intensive process instead.

Approach. Risk neutrality and $r_f = 0$ make every value an expected payoff. New equity is junior to the bond, so the most new shareholders can be offered is the whole residual claim $\mathbb{E}[\max\{X - 70, 0\}]$; the project is fundable iff that is at least \$75 million. This is the lecture's *debt overhang*: the bond captures part of the project's value. In 16.13 a riskier payoff shrinks the bond's share (*asset substitution*), which can make the equity issue feasible.

Notation.

- $F = 70$: existing senior bond obligation, with a covenant forbidding additional debt; $I = 75$: investment cost.
- States: recession (probability $\pi = 0.2$) and favourable (probability 0.8); X_s: project cash flow in state s; risk-neutral investors, $r_f = 0$, no taxes or bankruptcy costs.
- V: project value; $E^{\max}$: value of the whole residual claim after the bond, the most new shareholders can be offered; D: bond value.

Assumption. The firm has no other assets: without the project, the bond receives nothing. In 16.13 the investment still costs $I = 75$ (the statement changes only the cash flows).

Step 1 — (16.12) NPV of the project. We value the project's cash flows and subtract their cost:

$$\begin{aligned} V &= \pi X_{\text{rec}} + (1-\pi)X_{\text{fav}} \qquad (31)\\ &= (0.2)(85) + (0.8)(150)\\ &= 17 + 120 = 137,\\ NPV &= V - I = (137) - (75) = 62. \qquad (32)\end{aligned}$$

Step 2 — (16.12) Can it be funded?. We value the largest claim new shareholders can buy and compare it with the \$75 million they must pay. New debt is forbidden, so the money must come from equity, which is junior to the \$70 bond. Equity receives $\max\{85-70,0\} = 15$ in a recession and $\max\{150-70,0\} = 80$ in the favourable state:

$$\begin{aligned} E^{\max} &= (0.2)(15) + (0.8)(80) \qquad (33)\\ &= 3 + 64 = 67 < 75 = I.\end{aligned}$$

Even 100% of the equity is worth less than the \$75 million it must raise: **no**. The bond, which would be paid in full ($85 \geq 70$), would be worth $D = 70$ instead of 0; the split $70 + (67 - 75) = 62$ is the NPV. The creditors capture more than the whole NPV, which is why the shareholders cannot pay for it.

Answer (16.12). $NPV = \$62$ million, but the firm **cannot** fund the project: all of the equity is worth \$67 million < \$75 million.

Step 3 — (16.13) The capital-intensive variant. We repeat Steps 1–2 with the new cash flows, 5 (recession) and 170 (favourable):

$$\begin{aligned} V' &= (0.2)(5) + (0.8)(170)\\ &= 1 + 136 = 137,\\ NPV' &= (137) - (75) = 62,\end{aligned}$$

the same NPV but riskier. Equity receives $\max\{5-70,0\} = 0$ and $170 - 70 = 100$:

$$\begin{aligned} E'^{\max} &= (0.2)(0) + (0.8)(100) \qquad (34)\\ &= 80 \geq 75 = I.\end{aligned}$$

Yes: new shareholders pay \$75 million for a fraction $75/80 = 0.9375$ of the equity and the old shareholders keep $1 - 0.9375 = 6.25\%$, worth $0.0625(80) = 80 - 75 = 5$. The bond now receives $\min\{5, 70\} = 5$ and $\min\{170, 70\} = 70$, worth

$$\begin{aligned} D' &= (0.2)(5) + (0.8)(70)\\ &= 1 + 56 = 57,\end{aligned}$$

and $57 + 80 = 137 = V'$.

Answer (16.13). **Yes**: under the capital-intensive process the equity is worth \$80 million $\geq$ \$75 million, so new shareholders buy 93.75% of it for \$75 million.

Step 4 — Comparison. The two processes create the same value, 62. The risky one moves $70 - 57 = 13$ from the bondholders to the equity, which is exactly what lifts the equity's value from 67 to 80, above the \$75 million cost. The bondholders prefer the safe process, but under it nothing is built and they get 0, so they are better off with the risky one ($57 > 0$).

Intuition. Debt overhang: with a senior bond that cannot be diluted, new equity must pay for the whole project but receives only the value left after the bond, and here the bond absorbs too much of it. Raising risk shrinks the bond's share (asset substitution), and that makes the equity issue possible. Of the two distortions, the risk shift is the lesser here. A cleaner fix is to renegotiate the bond: under the safe process equity covers \$75 million whenever the face is cut to at most $137 - 75 = 62$, which the bondholders should accept since $62 > 57$ (the lecture's renegotiation argument).

Common mistakes.

- Answering "NPV > 0, so yes" in 16.12. Fundability depends on what the junior claim is worth to the new money, not on total value.
- Valuing equity as $V - F$. That gives $137 - 70 = 67$ for *both* processes and the wrong answer in 16.13; limited liability makes the recession payoff $\max\{5 - 70, 0\} = 0$, not -65.
- Forgetting that without the project the bond gets nothing here; that is why the bondholders prefer the risky process (57) to no project (0).

Ref: Grinblatt & Titman Ch. 16, § 16.2 "The Debt Overhang Problem" and "How the Seniority Structure of Debt Affects the Debt Overhang Problem", pp. 563–566 (Results 16.3–16.4); "The Asset Substitution Problem", pp. 569–572. Lecture Topic 4B, slides 19–31.

8. Project riskiness and credit rationing — Tirole 3.15, p. 150

What is asked. Two versions of the fixed-investment project have the same cost, the same expected income under good behaviour and the same effect of shirking on the success probability, but B succeeds less often with a larger payoff. Decide which version can be financed for a wider range of the entrepreneur's cash A, i.e. is less prone to credit rationing.

Approach. In the Holmström–Tirole fixed-investment model (lecture Topic 4A) a project is financed iff $A \geq \bar{A} = I - \mathcal{P}$, where $\mathcal{P} = p_H(R - B/\Delta p)$ is the *pledgeable income*. So the variant with the lower threshold $\bar{A}$, i.e. the higher pledgeable income, is less rationed. We write $\bar{A}$ as non-pledgeable rent minus NPV, use equal NPVs, and compare the rents.

Notation.

- $j \in \{A, B\}$: project variant; $p_H^j > p_L^j$: success probabilities with good / bad behaviour; $R^j > 0$: income on success (0 on failure).
- $\Delta p = p_H^A - p_L^A = p_H^B - p_L^B > 0$: common; $B > 0$: private benefit (common); I: common investment; A: borrower's cash.
- Maintained: $p_H^A R^A = p_H^B R^B$ (equal NPV) and $p_H^A > p_H^B$ (B riskier); (3.1), (3.2) hold for both variants.
- $\mathcal{P}^j = p_H^j(R^j - B/\Delta p)$: pledgeable income; $\bar{A}^j$: minimum cash for financing variant j (3.3).

Step 1 — The minimum stake is the same. We first derive the borrower's minimum reward in each variant. Variant j's incentive constraint is $p_H^j R_b^j \geq p_L^j R_b^j + B$, i.e. $(p_H^j - p_L^j)R_b^j \geq B$, i.e. $R_b^j \geq B/\Delta p$: the same for both variants, because B and Δp are common.

Step 2 — Financing thresholds. We compute each variant's threshold from (IR_l), $\mathcal{P}^j \geq I - A$. From (3.3),

$$\begin{aligned}\bar{A}^j &= I - \mathcal{P}^j \\ &= I - p_H^j \left(R^j - \frac{B}{\Delta p}\right) \\ &= I - p_H^j R^j + p_H^j \frac{B}{\Delta p} \\ &= p_H^j \frac{B}{\Delta p} - (p_H^j R^j - I). \end{aligned} \tag{35}$$

The first term is the non-pledgeable income: the expected rent the borrower must be left.

Step 3 — Compare. Subtracting (35) for B from that for A, and substituting $p_H^B R^B = p_H^A R^A$,

$$\begin{aligned}\bar{A}^A - \bar{A}^B &= \left[p_H^A \frac{B}{\Delta p} - (p_H^A R^A - I)\right] - \left[p_H^B \frac{B}{\Delta p} - ((p_H^A R^A) - I)\right] \\ &= p_H^A \frac{B}{\Delta p} - p_H^A R^A + I - p_H^B \frac{B}{\Delta p} + p_H^A R^A - I \\ &= (p_H^A - p_H^B)\frac{B}{\Delta p} > 0, \end{aligned} \tag{36}$$

positive because $p_H^A > p_H^B$, $B > 0$, $\Delta p > 0$. The riskier variant B is financed for every $A \geq \bar{A}^B$, including $A \in [\bar{A}^B, \bar{A}^A)$ where A is not: **variant B is less prone to credit rationing**.

Answer (3.15). Variant B, the riskier one: $\bar{A}^B = \bar{A}^A - (p_H^A - p_H^B)B/\Delta p < \bar{A}^A$, so every entrepreneur financed under A is financed under B, and those with $A \in [\bar{A}^B, \bar{A}^A)$ only under B (given that both variants need good behaviour, (3.2)).

Step 4 — Numerical illustration. $I = 100$, $\Delta p = 0.2$, $B = 9$; A: $p_H^A = 0.8$, $p_L^A = 0.6$, $R^A = 150$; B: $p_H^B = 0.6$, $p_L^B = 0.4$, $R^B = 200$. Both NPVs are $120 - 100 = 20$; (3.2) holds $(0.6(150) - 100 + 9 = -1 < 0$ and $0.4(200) - 100 + 9 = -11 < 0)$; $B/\Delta p = 45$. Then $\mathcal{P}^A = 0.8(105) = 84$, $\bar{A}^A = 16$; $\mathcal{P}^B = 0.6(155) = 93$, $\bar{A}^B = 7$; and $\bar{A}^A - \bar{A}^B = (0.2)(45) = 9$, as (36) says.

Step 5 — Beyond the maintained assumption (the book's extension). If (3.2) fails for the safe variant, a contract that leaves all income to investors ($R_b = 0$, the borrower misbehaves) may finance it: investors then receive $p_L^A R^A$, and the borrower, who gets the NPV with shirking, $p_L^A R^A + B - I$, when investors break even, wants it if that NPV is positive. The safe variant can then be financed and the risky one not when (in order) B cannot be financed with good behaviour, B cannot be financed with shirking, A can be financed with shirking, and A's NPV with shirking is positive:

$$I - A > p_H^B \left(R^B - \frac{B}{\Delta p}\right), \quad I - A > p_L^B R^B, \quad I - A \leq p_L^A R^A, \quad I < p_L^A R^A + B. \tag{37}$$

These conditions are compatible. With $A = 0$, $I = 120$, $\Delta p = 0.3$, $B = 45$, $p_H^A = 0.9$, $R^A = 200$, $p_H^B = 0.5$, $R^B = 360$ (both $p_H R = 180$): $p_H^B(R^B - B/\Delta p) = 0.5(210) = 105 < 120$; $p_L^B R^B = 0.2(360) = 72 < 120$; $p_L^A R^A = 0.6(200) = 120 \geq 120$; $p_L^A R^A + B = 165 > 120$. Here the risky variant is rationed and the safe one is funded. Good behaviour adds more pledgeable income when the success payoff is large, which favours the risky variant only when good behaviour is needed for funding.

Intuition. The borrower's rent $B/\Delta p$ is paid only on success. A safer project succeeds more often, so it pays out the rent more often and leaves less expected income to pledge: the non-pledgeable rent $p_H B/\Delta p$ is higher for the same NPV. Riskier projects with the same mean concentrate income in success, where it is pledged anyway.

Common mistakes.

- Importing the Stiglitz–Weiss intuition that riskier borrowers are rationed first. Here riskiness lowers the rent $p_H B/\Delta p$ and so *raises* pledgeable income.
- Computing the threshold as $I - p_H R$ (all income pledged), which is the same for both variants by $p_H^A R^A = p_H^B R^B$ and hides the difference.
- Forgetting that the comparison holds only while good behaviour is required for funding: if (3.2) fails for one variant, the book's extension (Step 5) can reverse the ranking.

Book answer. Matches the book's answer (p. 578): with the same minimum reward $B/\Delta p$, the safe project A is financed for a smaller range of A, i.e. it is more prone to credit rationing; the book then shows, with conditions (37), that the risky project may be the more rationed one if high-powered incentives are not necessarily called for. One slip in the book's prose: it says the third condition means that "the risky project" generates enough pledgeable income when investors get all the cash flow, but that condition, $I - A \leq p_L^A R^A$, is about the safe variant A.

Ref: Tirole Ch. 3, § 3.2.1–3.2.2 "The Fixed-Investment Model", "The Lenders' Credit Analysis", pp. 115–117. Lecture Topic 4A, slides 26–35 (determinants of credit rationing).

9. Entrepreneurial risk aversion — Tirole Exercise 3.2, p. 144

What is asked. (i) The entrepreneur is risk neutral above a subsistence level c_0 and infinitely averse to falling below it. Find the smallest cash $\bar{A}$ with which risk-neutral investors finance the project, and explain why the answer differs between a project that succeeds for sure under effort ($p_H = 1$) and one that can fail even under effort ($p_H < 1$). (ii) Redo the analysis for a general strictly concave utility, once with limited liability (her payments cannot be negative) and once without.

Approach. This is the fixed-investment model of the lecture (Topic 4A) with a risk-averse borrower. The contract pays her c^S on success and c^F on failure; she chooses it to maximise expected utility subject to the investors' breakeven constraint (IR_l) and her incentive constraint (IC_b), now written in utility. Pledgeable income is total expected income minus the cheapest expected payment that satisfies (IC_b) and the lower bound on c^F; a floor on consumption (subsistence in (i), limited liability in (ii)) raises that payment and so the minimum equity. In (ii) we write the Lagrangian and derive the first-order conditions, which give the Holmström–Shavell insurance-versus-incentives trade-off.

Notation.

- A: entrepreneur's cash ($A \geq c_0$ in (i)); $I > A$: fixed investment; $R > 0$: income in the case of success, 0 in the case of failure.
- p_H, $p_L = p_H - \Delta p$: success probabilities when she works / shirks ($0 < p_L < p_H \leq 1$); $B > 0$: private benefit from shirking.
- $NPV \equiv p_H R - I > 0$; $\bar{A}_{rn} \equiv I - p_H(R - B/\Delta p) > 0$: the risk-neutral threshold (3.3), positive by the assumption $I > p_H(R - B/\Delta p)$.
- c^S, c^F: the entrepreneur's consumption in the case of success / failure (her total payoff from the contract); $u(\cdot)$: utility of consumption; c_0: subsistence level in (i).
- $\mathcal{P}$: expected pledgeable income; $\bar{A}$: minimum cash for financing.
- $h(c) \equiv u^{-1}(u(c) + B/\Delta p)$: consumption in success that gives exactly the utility premium $B/\Delta p$ over consumption c in failure (part (ii)).

Assumption. The entrepreneur puts all of A into the project and the contract pays her c^S or c^F (keeping cash aside is the same as adding it to both payments). In (ii) the private benefit enters utility additively, $p\,u(c^S) + (1-p)\,u(c^F) + B$ when she shirks, as in the book's answer.

Step 1 — (i), $p_H < 1$: the programme. The entrepreneur offers the contract that maximises

her expected utility subject to the investors' breakeven and her incentive constraint:

$$\max_{c^S,c^F} \; p_H\,u(c^S) + (1-p_H)\,u(c^F) \tag{38}$$

$$\text{s.t.} \quad p_H(R-c^S) + (1-p_H)(0-c^F) \;\geq\; I - A, \tag{39}$$

$$p_H\,u(c^S) + (1-p_H)\,u(c^F) \;\geq\; p_L\,u(c^S) + (1-p_L)\,u(c^F) + B. \tag{40}$$

Failure has probability $1-p_H > 0$ even under work, so $c^F < c_0$ would make (38) equal to $-\infty$: the optimum has $c^F \geq c_0$, and then $c^S \geq c_0$ too. On $[c_0,\infty)$ utility is linear, $u(c)=c$, so (40) reads

$$\begin{aligned} p_H c^S + (1-p_H)c^F &\geq p_L c^S + (1-p_L)c^F + B \\ (p_H-p_L)c^S - \big[(1-p_L)-(1-p_H)\big]c^F &\geq B \\ (p_H-p_L)c^S - (p_H-p_L)c^F &\geq B \\ (\Delta p)(c^S-c^F) &\geq B. \end{aligned} \tag{41}$$

Step 2 — (i), $p_H < 1$: which constraints bind, and $\bar{A}$. With linear utility the objective (38) is the expected payment $p_H c^S + (1-p_H)c^F$. If (39) were slack, raising c^S would raise the objective and relax (41); so (39) binds and the entrepreneur gets $A + NPV$. Financing is possible if and only if the cheapest contract that satisfies (41) and $c^F \geq c_0$ leaves the investors enough. Using $c^S \geq c^F + B/\Delta p$ from (41) and then $c^F \geq c_0$, any feasible contract costs

$$\begin{aligned} p_H c^S + (1-p_H)c^F &\geq p_H\Big(c^F + \frac{B}{\Delta p}\Big) + (1-p_H)c^F \\ &= c^F + p_H\frac{B}{\Delta p} \;\geq\; c_0 + p_H\frac{B}{\Delta p}, \end{aligned}$$

with equality iff both inequalities bind, so the cheapest contract is

$$c^F = c_0, \tag{42}$$

$$c^S = c_0 + \frac{B}{\Delta p}. \tag{43}$$

The pledgeable income is total expected income minus this minimum expected payment; substituting (42)–(43),

$$\begin{aligned} \mathcal{P} &= p_H R - \Big[p_H\Big(c_0 + \frac{B}{\Delta p}\Big) + (1-p_H)(c_0)\Big] \\ &= p_H\Big(R - \frac{B}{\Delta p}\Big) - c_0. \end{aligned} \tag{44}$$

(the middle step: $p_H c_0 + (1-p_H)c_0 = c_0$). Financing requires $\mathcal{P} \geq I - A$; using (44) and moving terms,

$$\begin{aligned} p_H\Big(R - \frac{B}{\Delta p}\Big) - c_0 &\geq I - A \\ A \;\geq\; \bar{A} &= I + c_0 - p_H\Big(R - \frac{B}{\Delta p}\Big) \\ &= \Big[I - p_H\Big(R - \frac{B}{\Delta p}\Big)\Big] + c_0 = \bar{A}_{rn} + c_0. \end{aligned} \tag{45}$$

Because $\bar{A}_{rn} > 0$, $\bar{A} > c_0$: entrepreneurs with $A \in [c_0, \bar{A})$ are rationed. The participation constraint holds, since she gets $A + NPV > A$. Illustration: with $p_H = 0.8$, $p_L = 0.4$, $B = 1$, $R = 10$, $I = 6.5$, $c_0 = 1$, (45) gives $\bar{A} = (6.5) + (1) - (0.8)(10 - 1/0.4) = 1.5$, against $\bar{A}_{rn} = 0.5$.

Step 3 — (i), $p_H = 1$. Now failure never happens on the equilibrium path; it happens only if she shirks, with probability $1 - p_L > 0$. Set $c^F < c_0$ (for example $c^F = 0$, feasible under limited liability

when $c_0 > 0$). Shirking then yields $p_L\,u(c^S) + (1-p_L)(-\infty) + B = -\infty$, so (40) holds for every $c^S \geq c_0$ and the incentive problem costs nothing. The cheapest contract is $c^S = c_0$, and

$$\mathcal{P} = R - c_0, \tag{46}$$

$$\bar{A} = I + c_0 - R \;<\; c_0, \tag{47}$$

where the inequality uses $R - I = NPV > 0$. Every entrepreneur with $A \geq c_0$ is financed: no rationing (with the numbers above, $\bar{A} = 6.5 + 1 - 10 = -2.5$). The threshold is discontinuous at $p_H = 1$: by (45), $\bar{A} \to I + c_0 - R + B/\Delta p$ as $p_H \to 1$, a jump of $B/\Delta p$. If $c_0 \leq 0$, limited liability rules out a punishment below c_0 and the risk-neutral analysis applies.

Answer (i). For $p_H < 1$: $\bar{A} = I + c_0 - p_H(R - B/\Delta p) = \bar{A}_{rn} + c_0$, so entrepreneurs with $A \in [c_0, \bar{A})$ are rationed. For $p_H = 1$: failure reveals shirking and can be punished with $c^F < c_0$ at no cost on the equilibrium path, so $\bar{A} = I + c_0 - R < c_0$ and every entrepreneur with $A \geq c_0$ is financed (no rationing).

Step 4 — (ii) The general programme. With $u' > 0$, $u'' < 0$, the contract solves

$$\max_{c^S, c^F} \; p_H\,u(c^S) + (1-p_H)\,u(c^F) \tag{48}$$

$$\text{s.t.} \quad p_H(R - c^S) - (1-p_H)c^F \;\geq\; I - A, \tag{IR$_l$}$$

$$(\Delta p)\big[u(c^S) - u(c^F)\big] \;\geq\; B, \tag{IC$_b$}$$

$$c^F \geq 0,\; c^S \geq 0 \quad \text{(only under limited liability)}, \tag{LL}$$

$$p_H\,u(c^S) + (1-p_H)\,u(c^F) \;\geq\; u(A). \tag{PC}$$

(IC$_b$) is the risk-averse version of (IC$_b$): subtract $p_L u(c^S) + (1-p_L)u(c^F) + B$ from (48). (PC) says she prefers the project to consuming A. In utility units $v^S = u(c^S)$, $v^F = u(c^F)$ the objective and (IC$_b$) are linear and the cost $p_H u^{-1}(v^S) + (1-p_H)u^{-1}(v^F)$ is convex, so the problem is convex and the first-order conditions are sufficient.

Step 5 — (ii) Which constraints bind. We show both (IR$_l$) and (IC$_b$) bind, the second by deriving the first-order conditions. (a) (IR$_l$) binds: otherwise raise c^S, which raises (48) and relaxes (IC$_b$). (b) With multipliers $\lambda \geq 0$ on (IR$_l$) and $\mu \geq 0$ on (IC$_b$) (and (LL) slack), the Lagrangian is

$$\begin{aligned}\mathcal{L} = {} & p_H\,u(c^S) + (1-p_H)\,u(c^F) + \lambda\big[p_H(R - c^S) - (1-p_H)c^F - I + A\big] \\ & + \mu\big[(\Delta p)\big(u(c^S) - u(c^F)\big) - B\big].\end{aligned}$$

Differentiating with respect to c^S and c^F and setting the derivatives to zero,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c^S} &= p_H u'(c^S) - \lambda p_H + \mu(\Delta p)u'(c^S) = 0, \\ &\Longrightarrow \quad u'(c^S)(p_H + \mu\Delta p) = \lambda p_H, \\ \frac{\partial \mathcal{L}}{\partial c^F} &= (1-p_H)u'(c^F) - \lambda(1-p_H) - \mu(\Delta p)u'(c^F) = 0, \\ &\Longrightarrow \quad u'(c^F)(1 - p_H - \mu\Delta p) = \lambda(1-p_H).\end{aligned}$$

Here $\lambda > 0$, because $\lambda = 0$ would make the first condition read $u'(c^S)(p_H + \mu\Delta p) = 0$, impossible with $u' > 0$. Dividing the first by $\lambda p_H\,u'(c^S)$ and the second by $\lambda(1-p_H)\,u'(c^F)$,

$$\frac{1}{u'(c^S)} = \frac{1}{\lambda}\Big(1 + \mu\frac{\Delta p}{p_H}\Big), \tag{49}$$

$$\frac{1}{u'(c^F)} = \frac{1}{\lambda}\Big(1 - \mu\frac{\Delta p}{1-p_H}\Big). \tag{50}$$

If $\mu = 0$, (49)–(50) give $u'(c^S) = u'(c^F)$, i.e. full insurance $c^S = c^F$, which violates (IC$_b$) because $B > 0$. So $\mu > 0$ and, by (49)–(50), $c^S > c^F$: the entrepreneur bears risk, but only as much as incentives require (Holmström–Shavell inverse-marginal-utility rule).

Step 6 — (ii) The solution. We reduce the two binding constraints to one equation in c^F. The binding (IC_b) is $u(c^S) = u(c^F) + B/\Delta p$, i.e. $c^S = h(c^F)$. The binding (IR_l) is $p_H R - p_H c^S - (1 - p_H)c^F = I - A$, i.e.

$$p_H c^S + (1 - p_H)c^F = A + (p_H R - I) = A + NPV.$$

Substituting $c^S = h(c^F)$,

$$p_H\, h(c^F) + (1 - p_H)\, c^F = A + NPV. \tag{51}$$

Differentiating the identity $u(h(c)) = u(c) + B/\Delta p$ with respect to c,

$$u'(h(c))\, h'(c) = u'(c) \quad \Longrightarrow \quad h'(c) = \frac{u'(c)}{u'(h(c))} \;\geq\; 1,$$

because $h(c) > c$ and u' is decreasing. Hence the left-hand side of (51) is increasing in c^F, so (51) defines a unique $c^F = c(A)$, increasing in A. Substituting $u(c^S) = u(c^F) + B/\Delta p$ into (48), her utility is

$$\begin{aligned} V(A) &= p_H\Big(u(c(A)) + \frac{B}{\Delta p}\Big) + (1 - p_H)\, u(c(A)) \\ &= u(c(A)) + \frac{p_H B}{\Delta p}. \end{aligned} \tag{52}$$

Step 7 — (ii) Without limited liability. Only (PC) can fail. Let CE be the certainty equivalent of the lottery (c^S, c^F), $u(CE) = V(A)$, and $\pi \equiv \mathbb{E}[c] - CE$ its risk premium; by (51), $\mathbb{E}[c] = A + NPV$. Then

$$\text{financing} \iff V(A) \geq u(A) \iff CE \geq A \iff \pi(A) \leq NPV. \tag{53}$$

A risk-neutral entrepreneur ($\pi = 0$) without limited liability is always financed (she can be punished with $c^F < 0$; there is no $\bar{A}$). A risk-averse one is financed only if the risk premium of the incentive lottery does not exceed the NPV: risk aversion alone can block positive-NPV projects. Whether this gives a threshold in A depends on how π varies with wealth (with B in utility units, a richer entrepreneur needs a larger money spread to produce the utility gap $B/\Delta p$).

Step 8 — (ii) With limited liability. (LL) requires $c(A) \geq 0$. Because the left-hand side of (51) is increasing, $c(A) \geq 0$ iff $A + NPV \geq p_H h(0)$, i.e.

$$A \;\geq\; A_{LL} \equiv I - p_H\big[R - h(0)\big]. \tag{54}$$

Below A_{LL} no contract satisfies (IR_l), (IC_b) and (LL): the cheapest incentive-compatible contract, $c^F = 0$, $c^S = h(0)$, already costs more than $A + NPV$. Above A_{LL} the solution is that of Step 6 with $c^F = c(A) \geq 0$, so the minimum equity is $\bar{A} = \min\{A \geq A_{LL} :\; V(A) \geq u(A)\}$. Two remarks. First, for $A > A_{LL}$ the entrepreneur receives $c^F > 0$ in failure: unlike the risk-neutral case, pure debt ($c^F = 0$) is no longer optimal, because she buys partial insurance. Second, risk neutrality gives back the book: with $u(c) = c$, $h(0) = B/\Delta p$ and (54) is (45) with $c_0 = 0$; part (i) is the case of a floor c_0 instead of 0. If u is normalised so that $u'(0) = 1$, concavity gives $h(0) \geq B/\Delta p$, so $A_{LL} \geq \bar{A}_{rn}$: risk aversion raises the equity requirement.

Answer (ii). Both (IR_l) and (IC_b) bind; the contract solves (51) with $u(c^S) = u(c^F) + B/\Delta p$, and the first-order conditions (49)–(50) give $c^S > c^F$ (partial insurance). Without limited liability, financing obtains iff the risk premium of the incentive lottery is at most the NPV, $\pi(A) \leq NPV$. With limited liability, also $A \geq A_{LL} = I - p_H[R - h(0)]$, and $\bar{A} = \min\{A \geq A_{LL} : V(A) \geq u(A)\}$; above A_{LL} she gets $c^F > 0$, so pure debt is no longer optimal.

Intuition. Incentives need a spread between the success and failure payoffs. A risk-neutral entrepreneur bears that spread for free; a subsistence floor or limited liability bounds the punishment,

so the spread must be paid as a rent in success, and a risk-averse entrepreneur must in addition be compensated for the risk. Both raise the stake she keeps and lower pledgeable income. The $p_H = 1$ case is the exception: failure then proves shirking, so an unbounded punishment off the equilibrium path removes the agency cost.

Common mistakes.

- Subtracting c_0 from the cash instead of adding it to the payments: the entrepreneur must be paid at least c_0 in *both* states, which costs investors c_0 in expectation; the two bookkeepings agree only if this is done consistently.
- Treating $p_H = 1$ like $p_H < 1$. With $p_H = 1$ failure never occurs under effort, so the $-\infty$ punishment is free; with $p_H < 1$ it would hit her on the equilibrium path.
- In (ii), carrying over the risk-neutral result that pure debt ($c^F = 0$) is optimal. A risk-averse borrower buys partial insurance, $c^F > 0$, whenever $A > A_{LL}$.

Book answer. Matches the book's answer (p. 571): $\bar{A} = I + c_0 - p_H(R - B/\Delta p)$ for $p_H < 1$, the programme of (ii) with binding incentive constraint. For $p_H = 1$ the book writes the pledgeable income as $p_H R$, i.e. with c_0 kept aside from A; this is (46)–(47) in other bookkeeping. The book only states the programme for (ii) and cites Holmström (1979) and Shavell (1979); Steps 6–8 are this solution's.

Ref: Tirole §3.2.1–3.2.2, pp. 115–117; answer p. 571.

10. Maximal incentives principle in the fixed-investment model — Tirole Exercise 3.18, p. 151

What is asked. In the fixed-investment model, failure now yields a positive income R^F (salvage value) instead of 0. Show that a debt contract (investors take everything in failure and a fixed amount in success) is an optimal financial structure, and show that, unlike the variable-investment model of Section 3.4.3, other contracts may be optimal too.

Approach. This is the lecture's "debt dominates equity" result (maximal incentives principle, Topic 4A). Set up the borrower's programme: maximise her expected reward subject to (IC_b), (IR_l) and limited liability. With a fixed investment her utility equals the NPV for *every* contract on which (IR_l) binds, so the contract matters only for feasibility, i.e. for *pledgeable income*. Paying her in failure buys no incentive, so the contract that maximises pledgeable income pays her nothing in failure: debt. Uniqueness fails because, when $A > \bar{A}$, the spare pledgeable income can be paid in either state.

Notation.

- I: fixed investment; A: entrepreneur's cash; $R^S > R^F > 0$: income in success / failure, with $R^F < I - A$; $R \equiv R^S - R^F$.
- p_H, $p_L = p_H - \Delta p$, B: as in Section 3.2.
- R_b^S, $R_b^F \geq 0$: entrepreneur's reward in success / failure (limited liability).
- $NPV \equiv p_H R^S + (1 - p_H)R^F - I = p_H R + R^F - I > 0$; U_b: entrepreneur's net utility.
- D: face value of debt; $\bar{A}$: minimum cash for financing.

Step 1 — Programme. The borrower designs the contract (R_b^S, R_b^F); her incentive constraint is $p_H R_b^S + (1 - p_H)R_b^F \geq p_L R_b^S + (1 - p_L)R_b^F + B$, which simplifies (collect R_b^S and R_b^F terms,

$(1-p_L)-(1-p_H)=\Delta p$) to (IC_b):

$$U_b = \max_{R_b^S, R_b^F} p_H R_b^S + (1-p_H)R_b^F - A \tag{55}$$
$$\text{s.t.} \quad (\Delta p)(R_b^S - R_b^F) \;\geq\; B, \tag{IC_b}$$
$$p_H(R^S - R_b^S) + (1-p_H)(R^F - R_b^F) \;\geq\; I - A, \tag{IR_l}$$
$$R_b^S \geq 0,\; R_b^F \geq 0. \tag{LL}$$

Step 2 — The breakeven constraint binds; U_b is the NPV. If (IR_l) were slack, raising R_b^S would raise (55) and relax (IC_b). With (IR_l) binding, solve it for her expected reward,

$$p_H R^S - p_H R_b^S + (1-p_H)R^F - (1-p_H)R_b^F = I - A$$
$$p_H R_b^S + (1-p_H)R_b^F = p_H R^S + (1-p_H)R^F - (I-A),$$

and substitute it into (55):

$$\begin{aligned} U_b &= \left[p_H R^S + (1-p_H)R^F - (I-A)\right] - A \\ &= p_H R^S + (1-p_H)R^F - I \\ &= NPV. \end{aligned} \tag{56}$$

Because I is fixed, (56) is the same for *every* feasible contract with binding (IR_l). The financial structure matters only for feasibility: it must deliver (IC_b) while leaving investors $I-A$.

Step 3 — Maximal pledgeable income and the financing condition. The constraint set is nonempty iff the largest expected income that can go to investors subject to (IC_b) and (LL) covers $I-A$, i.e. iff expected income minus the *smallest* expected reward compatible with (IC_b) and (LL) is at least $I-A$. Using $R_b^S \geq R_b^F + B/\Delta p$ from (IC_b),

$$\begin{aligned} p_H R_b^S + (1-p_H)R_b^F &\geq p_H\Big(R_b^F + \frac{B}{\Delta p}\Big) + (1-p_H)R_b^F \\ &= R_b^F + p_H\frac{B}{\Delta p} \;\geq\; p_H\frac{B}{\Delta p}, \end{aligned}$$

the last step by $R_b^F \geq 0$; both bounds are attained by $R_b^F = 0$, $R_b^S = B/\Delta p$. Each unit of failure reward costs one unit of expected income and buys no incentive. So

$$\begin{aligned} \mathcal{P} &= p_H R^S + (1-p_H)R^F - p_H\frac{B}{\Delta p} \\ &= p_H(R^S - R^F) + R^F - p_H\frac{B}{\Delta p} \\ &= R^F + p_H\Big(R - \frac{B}{\Delta p}\Big), \end{aligned} \tag{57}$$

and financing obtains iff $\mathcal{P} \geq I-A$, i.e.

$$A \;\geq\; \bar{A} \equiv I - R^F - p_H\Big(R - \frac{B}{\Delta p}\Big) = p_H\frac{B}{\Delta p} - NPV, \tag{58}$$

where the second form uses $NPV = p_H R + R^F - I$. The safe income R^F works exactly like cash on hand.

Step 4 — A debt contract is optimal. Let investors hold debt of face value D defined by

$$p_H D + (1-p_H)R^F = I - A. \tag{59}$$

In failure the firm defaults and investors take R^F; in success they get D. By $R^F < I - A$, (59) gives $D > R^F$, so the borrower gets $R_b^F = 0$. Substituting (59) into (58) (written as $p_H(R^S - B/\Delta p) + (1 - p_H)R^F \geq I - A$),

$$\begin{aligned} p_H\Big(R^S - \frac{B}{\Delta p}\Big) + (1-p_H)R^F &\geq p_H D + (1-p_H)R^F \\ p_H\Big(R^S - \frac{B}{\Delta p}\Big) &\geq p_H D \\ R^S - \frac{B}{\Delta p} &\geq D \\ R_b^S = R^S - D &\geq \frac{B}{\Delta p}, \end{aligned} \tag{60}$$

(subtracting $(1 - p_H)R^F$, then dividing by $p_H > 0$, then rearranging) so (IC_b) holds, (IR_l) binds by construction, and by (56) debt attains the maximum. It is optimal whenever financing is feasible.

Step 5 — Debt is not uniquely optimal. The optimal set is every $(R_b^S, R_b^F) \geq 0$ with binding (IR_l) and (IC_b). From the binding (IR_l), $R_b^S = [A + NPV - (1 - p_H)R_b^F]/p_H$. Substituting this into (IC_b),

$$\begin{aligned} (\Delta p)\Big(\frac{A + NPV - (1-p_H)R_b^F}{p_H} - R_b^F\Big) &\geq B \\ (\Delta p)\,\frac{A + NPV - (1-p_H)R_b^F - p_H R_b^F}{p_H} &\geq B \\ \frac{A + NPV - R_b^F}{p_H} &\geq \frac{B}{\Delta p} \\ A + NPV - R_b^F &\geq p_H \frac{B}{\Delta p} \\ R_b^F &\leq A + NPV - p_H\frac{B}{\Delta p} = A - \bar{A}, \end{aligned} \tag{61}$$

using (58) in the last line. So every $R_b^F \in [0, A - \bar{A}]$ is optimal (add $R_b^F \leq R^F$ if investors' claim must be nonnegative in each state). The contract is unique only at $A = \bar{A}$. For $A > \bar{A}$, a small failure reward (debt plus a little equity, or even all-equity when its implied R_b^F satisfies (61)) does as well. Illustration: $R^S = 12$, $R^F = 2$, $p_H = 0.8$, $p_L = 0.4$, $B = 1.6$, $I = 9$, $A = 3$. Then $B/\Delta p = 1.6/0.4 = 4$, $NPV = 0.8(12) + 0.2(2) - 9 = 9.6 + 0.4 - 9 = 1$, $\bar{A} = 0.8(4) - 1 = 2.2$, and by (59) $D = (9 - 3 - 0.2(2))/0.8 = 5.6/0.8 = 7$ (with $2 < 7 \leq 12 - 4 = 8$). The optimal set is $R_b^F \in [0,\, 3 - 2.2] = [0, 0.8]$; at $R_b^F = 0.8$, $R_b^S = (3 + 1 - 0.2(0.8))/0.8 = 3.84/0.8 = 4.8$ and (IC_b) binds: $0.4(4.8 - 0.8) = 1.6 = B$.

Step 6 — Contrast with variable investment. In Section 3.4.3 the entrepreneur's utility $(p_H R + R^F - 1)I$ rises with I, so she expands I until the financing constraint binds, the analogue of $A = \bar{A}$. Then the slack in (61) is zero, $R_b^F = 0$ is forced and debt is the unique optimum; any $R_b^F > 0$ would waste incentive power and reduce the borrowing capacity (3.16).

Answer (3.18). Debt with face value D given by $p_H D + (1 - p_H)R^F = I - A$ (so $R^F < D \leq R^S - B/\Delta p$) is optimal whenever the project can be financed, $A \geq \bar{A} = I - R^F - p_H(R - B/\Delta p)$. It is not uniquely optimal: every contract with $R_b^F \in [0, A - \bar{A}]$ and binding (IR_l) gives the same utility NPV; the optimum is unique only at $A = \bar{A}$.

Intuition. Paying the entrepreneur in failure costs expected income but weakens incentives, so it is the worst way to reward her; debt, which gives her nothing in failure, maximises incentives per unit of pay. With a fixed project, once the project is funded there is no further use for pledgeable income, so any leftover slack can be paid out in either state.

Common mistakes.

- Concluding that all-equity is never optimal, as in Section 3.4.3. That needs the investment to expand until the financing constraint binds; with a fixed I and $A > \bar{A}$ it does not bind.
- Leaving R^F out of pledgeable income. The salvage value is not subject to moral hazard and works exactly like cash on hand: $\bar{A}$ falls one for one with R^F.
- Writing (IC_b) as $(\Delta p)R_b^S \geq B$, forgetting that a failure reward R_b^F weakens incentives: what matters is the spread $R_b^S - R_b^F$.

Book answer. Matches the book's answer (p. 579), including condition (1) there, $p_H(R - B/\Delta p) \geq I - (A + R^F)$, the debt implementation with $R^F < D < R^S$ and the non-uniqueness when (IR_l) holds with slack at $R_b^F = 0$; (61) makes the size of the optimal set explicit.

Ref: Tirole §3.2.2, pp. 116–117; §3.4.3, pp. 128–129; answer p. 579.

11. Retained-earnings benefit — Tirole Exercise 3.11, p. 148

What is asked. (i) Find how much one extra unit of retained earnings (the entrepreneur's date-1 profit kept in the firm) is worth to her at date 2, given that it helps finance a second project whose private benefit B^2 is still random. (ii) Show, by argument or example, that the existence of the second project can get the first project financed even though no date-2 income can be pledged at date 1.

Approach. Two applications of the fixed-investment model (lecture Topic 4A, investment and internal funds). At date 2, retained earnings play the role of cash A in (IR_l): more of them lets the second project be financed for higher values of B^2. Their *shadow value* is the derivative of the entrepreneur's gross date-2 utility. At date 1, that continuation value is lost when project 1 fails, so it works as an extra reward for effort: the date-1 incentive constraint needs less cash reward, and date-1 *pledgeable income* rises.

Notation.

- Project $t \in \{1, 2\}$: investment I^t, income R^t in success, probabilities p_H^t, $p_L^t = p_H^t - \Delta p^t$, private benefit B^t; $NPV^2 \equiv p_H^2 R^2 - I^2 > 0$.
- A^1: entrepreneur's cash at date 1; R_b^1: her date-1 reward in success (her retained earnings), 0 in failure.
- $B^2 \in [\underline{B}^2, \bar{B}^2]$: random, c.d.f. F, density f, revealed at the start of date 2.
- $\hat{B}^2(R)$: largest B^2 for which project 2 is financed with retained earnings R.
- $W(R)$: entrepreneur's expected gross date-2 utility with retained earnings R; $\mu(R) \equiv W'(R)$: shadow value of retained earnings (v in §3.4.2).

Assumption. No discounting; the entrepreneur's only date-2 wealth is R_b^1 (or 0), and date-2 lenders cannot be pledged date-1 income.

Step 1 — (i) Date-2 financing given retained earnings R. Project 2 is financed iff its pledgeable income covers the outside funding, (IR_l) for project 2:

$$p_H^2\Big(R^2 - \frac{B^2}{\Delta p^2}\Big) \geq I^2 - R. \tag{62}$$

Solving (62) for B^2 (expand, move $p_H^2 B^2/\Delta p^2$ to the right and $I^2 - R$ to the left, then multiply by $\Delta p^2/p_H^2 > 0$),

$$\begin{aligned} p_H^2 R^2 - \frac{p_H^2 B^2}{\Delta p^2} &\geq I^2 - R \\ \frac{p_H^2 B^2}{\Delta p^2} &\leq p_H^2 R^2 - I^2 + R = NPV^2 + R \\ B^2 \;\leq\; \hat{B}^2(R) &\equiv \frac{\Delta p^2}{p_H^2}(NPV^2 + R). \end{aligned} \tag{63}$$

By the assumption $\underline{B}^2 > \Delta p^2\, NPV^2/p_H^2$, (63) gives $\hat{B}^2(0) < \underline{B}^2$: with no retained earnings, project 2 is never financed.

Step 2 — (i) Gross date-2 utility. We follow the hint and compute her expected gross utility. If financed, project 2's lenders break even and the entrepreneur gets her cash plus NPV^2 (equation (3.6)); otherwise she keeps R. With probability $F(\hat{B}^2(R))$ it is financed, so

$$\begin{aligned} W(R) &= F\big(\hat{B}^2(R)\big)\big(R + NPV^2\big) + \Big[1 - F\big(\hat{B}^2(R)\big)\Big]R \\ &= R + F\big(\hat{B}^2(R)\big)\, NPV^2. \end{aligned} \tag{64}$$

Step 3 — (i) Shadow value. Differentiating (64) and using $d\hat{B}^2/dR = \Delta p^2/p_H^2$ from (63),

$$\begin{aligned} \mu(R) = W'(R) &= 1 + f\big(\hat{B}^2(R)\big)\,\frac{d\hat{B}^2}{dR}\, NPV^2 \\ &= 1 + f\big(\hat{B}^2(R)\big)\,\frac{\Delta p^2}{p_H^2}\, NPV^2 \;\geq 1. \end{aligned} \tag{65}$$

One unit of retained earnings is worth one unit of consumption plus the expected NPV it unlocks: it raises the probability of financing at the margin by $f\,\Delta p^2/p_H^2$.

Answer (i). $\mu(R) = 1 + f\big(\hat{B}^2(R)\big)\,(\Delta p^2/p_H^2)\,(p_H^2 R^2 - I^2) \;\geq 1$, with $\hat{B}^2(R) = (\Delta p^2/p_H^2)(p_H^2 R^2 - I^2 + R)$ and gross utility $W(R) = R + F(\hat{B}^2(R))(p_H^2 R^2 - I^2)$.

Step 4 — (ii) Date-1 incentive constraint. By (64) and Step 1, $W(0) = 0 + F(\hat{B}^2(0))\, NPV^2 = 0$, so the entrepreneur works at date 1 iff

$$\begin{aligned} p_H^1 W(R_b^1) + (1 - p_H^1)W(0) &\geq p_L^1 W(R_b^1) + (1 - p_L^1)W(0) + B^1 \\ (p_H^1 - p_L^1)W(R_b^1) - (p_H^1 - p_L^1)W(0) &\geq B^1 \\ (\Delta p^1)\, W(R_b^1) &\geq B^1 \qquad (W(0) = 0) \\ (\Delta p^1)\big[R_b^1 + F(\hat{B}^2(R_b^1))NPV^2\big] &\geq B^1, \end{aligned} \tag{IC^1}$$

the last line substituting (64). Let $\tilde{R}_b^1$ be the smallest R_b^1 satisfying (IC^1). Whenever $F(\hat{B}^2(\tilde{R}_b^1)) > 0$,

$$\tilde{R}_b^1 = \frac{B^1}{\Delta p^1} - F\big(\hat{B}^2(\tilde{R}_b^1)\big)NPV^2 < \frac{B^1}{\Delta p^1}. \tag{66}$$

Step 5 — (ii) Date-1 financing. The entrepreneur maximises $p_H^1 W(R_b^1) - A^1$ subject to (IC^1) and $p_H^1(R^1 - R_b^1) \geq I^1 - A^1$. Since W is increasing (65), the breakeven constraint binds, and financing is possible iff

$$p_H^1(R^1 - \tilde{R}_b^1) \;\geq\; I^1 - A^1. \tag{67}$$

By (66) the left-hand side exceeds the stand-alone pledgeable income $p_H^1(R^1 - B^1/\Delta p^1)$. So for A^1 with $p_H^1(R^1 - B^1/\Delta p^1) < I^1 - A^1 \leq p_H^1(R^1 - \tilde{R}_b^1)$, project 1 is funded only because project 2 exists, although none of project 2's income is pledged at date 1.

Step 6 — (ii) An example. Project 2: $I^2 = 20$, $R^2 = 40$, $p_H^2 = 0.75$, $p_L^2 = 0.25$, $B^2 \sim U[7, 9]$, so $NPV^2 = 10$ and $\Delta p^2 NPV^2/p_H^2 = 20/3 < 7$ (assumption holds; also $p_L^2 R^2 + B^2 < I^2$). By (63), $\hat{B}^2(R) = \frac{2}{3}(10 + R)$, which lies in $[7, 9]$ for $R \in [0.5, 3.5]$; there $F = (\hat{B}^2 - 7)/2$ and (64)–(65) give

$$\begin{aligned} W(R) &= R + \frac{\frac{2}{3}(10 + R) - 7}{2}\,(10) \\ &= R + \frac{(20 + 2R - 21)/3}{2}\,(10) = R + \frac{5(2R - 1)}{3} \\ &= \frac{3R + 10R - 5}{3} = \frac{13R - 5}{3}, \\ \mu &= 1 + \Big(\frac{1}{2}\Big)\Big(\frac{0.5}{0.75}\Big)(10) = 1 + \frac{10}{3} = \frac{13}{3}. \end{aligned}$$

Project 1: $I^1 = 4.5$, $R^1 = 10$, $p_H^1 = 0.5$, $p_L^1 = 0.25$, $B^1 = 0.5$, so $B^1/\Delta p^1 = 2$. Alone, the pledgeable income is $(0.5)(10-2) = 4$ and $\bar{A}^1 = 4.5 - 4 = 0.5$. With project 2, ($IC^1$) at equality reads $W(\tilde{R}_b^1) = B^1/\Delta p^1$, i.e. $(13\tilde{R}_b^1 - 5)/3 = 2$, so $13\tilde{R}_b^1 = 6 + 5 = 11$, $\tilde{R}_b^1 = 11/13 \approx 0.846154$ (inside $[0.5, 3.5]$, where the formula for W holds), and

$$p_H^1(R^1 - \tilde{R}_b^1) = (0.5)\Big(10 - \frac{11}{13}\Big) = \frac{119}{26} \approx 4.576923 > 4.5.$$

The threshold falls from 0.5 to $4.5 - 119/26 = -1/13 \approx -0.076923$: any $A^1 \in [0, 0.5)$ is now funded (at $A^1 = 0$, $R_b^1 = 10 - 4.5/0.5 = 1 \geq \tilde{R}_b^1$).

Answer (ii). Yes. Because project 2's rent is lost on failure, the date-1 reward needed for effort falls to $\tilde{R}_b^1 < B^1/\Delta p^1$ and date-1 pledgeable income rises to $p_H^1(R^1 - \tilde{R}_b^1) > p_H^1(R^1 - B^1/\Delta p^1)$. Any A^1 with $p_H^1(R^1 - B^1/\Delta p^1) < I^1 - A^1 \leq p_H^1(R^1 - \tilde{R}_b^1)$ is financed only thanks to project 2 (in the example, $A^1 \in [0, 0.5)$).

Intuition. Retained earnings are worth more than their face value because they are the net worth that gets the next project financed. That future rent is lost if project 1 fails, so it acts as a reward for date-1 effort: part of the incentive is paid with continuation value instead of cash, and the cash saved becomes pledgeable to date-1 lenders.

Common mistakes.

- Valuing retained earnings at face ($\mu = 1$). Each unit also raises the chance that project 2 is financed, worth $f\,(\Delta p^2/p_H^2)\,NPV^2$ more.
- Writing the date-1 incentive constraint with R_b^1 alone, $(\Delta p^1)R_b^1 \geq B^1$: the reward for success is $W(R_b^1)$, which includes the continuation rent.
- Treating project 2's income as pledgeable at date 1. The effect works only through the entrepreneur's incentives; date-1 lenders are repaid out of R^1.

Book answer. Matches the book's answer (p. 575): $\mu = 1 + f(\hat{B}^2)(d\hat{B}^2/dR_b^1)NPV^2$ and the date-1 constraint $(\Delta p^1)[R_b^1 + F(\hat{B}^2(R_b^1))NPV^2] \geq B^1$. The book writes the constraint without $W(0)$; this is right because the assumption on $\underline{B}^2$ makes $W(0) = 0$ (Step 1). The numerical example is this solution's.

Ref: Tirole §3.2.2, pp. 116–117; §3.4.2, p. 128 (shadow value of equity); answer p. 575.

12. Renegotiation and debt forgiveness — Tirole Exercise 3.6, p. 146

What is asked. In the variable-investment model, suppose the initial loan leaves the borrower too small a stake to work, $R_b < BI/\Delta p$, but can be renegotiated after the investment and before effort. (i) Show that a mutually beneficial renegotiation exists exactly when the stated inequality holds. (ii) Explain that inequality, and derive it as the financing condition of a fictitious fixed-investment project. (iii) If the borrower makes a take-it-or-leave-it renegotiation offer, find how much she can invest (her borrowing capacity) with such an initial contract. (iv) Argue directly that writing $R_b \geq BI/\Delta p$ from the start loses nothing.

Approach. Lecture Topic 4B, renegotiation. Without renegotiation the borrower shirks; renegotiation can only help if it restores (IC_b), and it happens iff both sides gain relative to the shirking status quo. The lender gives up part of his claim (*debt forgiveness*) and is paid back out of the extra *pledgeable income* that effort creates, so the condition has the form "pledgeable income $\geq$ investment $-$ cash" of the fixed-investment model. For (iii) the lender's ex ante breakeven uses his renegotiated payoff; for (iv) rational expectations make the renegotiated contract one that could have been signed directly.

Notation.

- I: (sunk) investment; income RI in success, 0 in failure; A: borrower's cash.

- p_H, $p_L = p_H - \Delta p$; private benefit BI from shirking; (3.7) $p_H R > 1$, (3.8) $1 > p_L R + B$, (3.9) $p_H R < 1 + p_H B/\Delta p$; these imply $(\Delta p)R > B$.
- $R_b < BI/\Delta p$: borrower's stake in the initial contract; $\hat{R}_b$: renegotiated stake.
- $k = 1/[1 - p_H(R - B/\Delta p)]$: equity multiplier (3.12).

Step 1 — (i) Status quo. We compute each party's payoff if nothing is renegotiated; these are the outside options in the renegotiation. With $R_b < BI/\Delta p$, $(\Delta p)R_b < BI$: (IC_b) fails and the borrower shirks (success probability p_L, private benefit BI). Payoffs if the contract is not renegotiated:

$$U_b^0 = p_L R_b + BI, \tag{68}$$

$$U_l^0 = p_L(RI - R_b). \tag{69}$$

Step 2 — (i) Renegotiation must restore incentives. A new stake $\hat{R}_b < BI/\Delta p$ still induces shirking, so total surplus $p_L RI + BI$ is unchanged and the change is a pure transfer: one side loses. A mutually advantageous renegotiation therefore needs $\hat{R}_b \geq BI/\Delta p$, and both parties must gain relative to (68)–(69):

$$p_H \hat{R}_b \geq p_L R_b + BI, \tag{70}$$

$$p_H(RI - \hat{R}_b) \geq p_L(RI - R_b). \tag{71}$$

Step 3 — (i) The condition. For any $\hat{R}_b \geq BI/\Delta p$, (70) holds automatically, since $R_b < BI/\Delta p$:

$$p_H \hat{R}_b \geq p_H \frac{BI}{\Delta p} = BI + p_L \frac{BI}{\Delta p} > BI + p_L R_b.$$

The lender prefers the smallest admissible stake, $\hat{R}_b = BI/\Delta p$. So a mutually advantageous $\hat{R}_b$ exists iff (71) holds at $\hat{R}_b = BI/\Delta p$:

$$\begin{aligned} p_H\Big(RI - \frac{BI}{\Delta p}\Big) &\geq p_L(RI - R_b) \\ p_H RI - \frac{p_H BI}{\Delta p} &\geq p_L RI - p_L R_b \\ (p_H - p_L)RI - \frac{p_H BI}{\Delta p} + p_L R_b &\geq 0 \\ (\Delta p)RI - \frac{p_H BI}{\Delta p} + p_L R_b &\geq 0. \end{aligned} \tag{72}$$

Necessity: any renegotiated contract has $\hat{R}_b \geq BI/\Delta p$, and (71) is hardest to meet at larger $\hat{R}_b$. Sufficiency: if (72) holds, $\hat{R}_b = BI/\Delta p$ makes the lender weakly and the borrower strictly better off. The renegotiation is *debt forgiveness*: the lender's claim in success falls from $RI - R_b$ to $RI - \hat{R}_b$.

Answer (i). Renegotiation (to $\hat{R}_b \geq BI/\Delta p$) takes place iff $(\Delta p)RI - p_H BI/\Delta p + p_L R_b \geq 0$.

Step 4 — (ii) Interpretation via the general theory. Treat "inducing effort" as a fictitious fixed-investment project. Its income is the increase in expected output, $(\Delta p)RI$; it needs no new investment ($I_f = 0$); the borrower's minimum stake in it is $p_H\, BI/\Delta p$ (her expected rent under (IC_b)), and she brings as "cash" the expected income she gives up, $A_f \equiv p_L R_b$. Its pledgeable income is $(\Delta p)RI - p_H BI/\Delta p$, and the financing condition (IR_l), "pledgeable income $\geq$ investment $-$ cash", reads

$$(\Delta p)RI - \frac{p_H BI}{\Delta p} \;\geq\; 0 - p_L R_b, \tag{73}$$

which is (72). The total gain from restoring effort, $(\Delta p)RI - BI$, is always positive, yet renegotiation may fail: the lender can only be paid out of pledgeable income, and the borrower's initial stake $p_L R_b$ plays the role of net worth. A larger R_b makes forgiveness easier.

Answer (ii). The condition says the pledgeable part of the effort gain, $(\Delta p)RI - p_H BI/\Delta p$, must cover the lender's "outlay" $0 - p_L R_b$: it is (IR$_l$) for a fictitious project with income $(\Delta p)RI$, investment 0 and cash $p_L R_b$.

Step 5 — (iii) Borrowing capacity under renegotiation, borrower has all bargaining power. When (72) holds, the borrower offers the $\hat{R}_b$ that leaves the lender indifferent:

$$p_H(RI - \hat{R}_b) = p_L(RI - R_b), \quad \text{i.e.} \quad \hat{R}_b = RI - \frac{p_L}{p_H}(RI - R_b). \tag{74}$$

The lender therefore anticipates exactly (69), and his ex ante breakeven constraint is

$$p_L(RI - R_b) \;\geq\; I - A. \tag{75}$$

The borrower's utility is total surplus with effort minus the lender's payoff: $p_H RI - p_L(RI - R_b) - A$, which by the binding (75) equals $p_H RI - (I - A) - A = (p_H R - 1)I$, the NPV with effort. Since $p_H R > 1$ (3.7), she maximises I subject to (72), (75) and $0 \leq R_b < BI/\Delta p$. (72) bounds $p_L R_b$ from below and (75) (expanded: $p_L RI - p_L R_b \geq I - A$) bounds it from above:

$$\frac{p_H BI}{\Delta p} - (\Delta p)RI \;\leq\; p_L R_b \;\leq\; p_L RI - (I - A). \tag{76}$$

An R_b exists iff the lower bound is below the upper bound:

$$\begin{aligned}
\frac{p_H BI}{\Delta p} - (\Delta p)RI &\leq p_L RI - I + A \\
I - (\Delta p)RI - p_L RI + \frac{p_H BI}{\Delta p} &\leq A \\
I - p_H RI + \frac{p_H BI}{\Delta p} &\leq A \qquad (\Delta p + p_L = p_H) \\
I\Big[1 - p_H\Big(R - \frac{B}{\Delta p}\Big)\Big] &\leq A \\
I &\leq kA.
\end{aligned} \tag{77}$$

If $p_H B/\Delta p \geq (\Delta p)R$, the lower bound in (76) is nonnegative and the capacity is exactly kA: at $I = kA$ both (72) and (75) bind, $R_b < BI/\Delta p$ (by $(\Delta p)R > B$) and (74) gives $\hat{R}_b = BI/\Delta p$. If $p_H B/\Delta p < (\Delta p)R$, the constraint $R_b \geq 0$ binds first, and (75) with $R_b = 0$ gives $I \leq A/(1 - p_L R) < kA$. Either way the capacity is at most kA, the capacity without renegotiation.

Answer (iii). The borrowing capacity is $I = kA$, $k = 1/[1 - p_H(R - B/\Delta p)]$, when $p_H B/\Delta p \geq (\Delta p)R$ (then $p_L R_b = kA[p_H B/\Delta p - (\Delta p)R]$ and $\hat{R}_b = BI/\Delta p$), and $I = A/(1 - p_L R) < kA$ otherwise; it never exceeds the capacity kA of an incentive-compatible initial contract.

Step 6 — (iv) Rational expectations. Take any initial contract (I, R_b) with $R_b < BI/\Delta p$. If it is not renegotiated, the borrower shirks; the NPV is $(p_L R + B - 1)I < 0$ by (3.8), and since the lender breaks even the borrower's utility is that negative NPV: worse than not investing. If it is renegotiated to $\hat{R}_b \geq BI/\Delta p$, the final contract $(I, \hat{R}_b)$ satisfies (IC$_b$), and by rational expectations the lender's ex ante breakeven uses his final payoff, $p_H(RI - \hat{R}_b) \geq I - A$, which is (IR$_l$). So $(I, \hat{R}_b)$ could have been signed directly, with the same effort and the same payoffs. Nothing is lost by restricting attention to $R_b \geq BI/\Delta p$, which is renegotiation-proof.

Answer (iv). Any initial contract with $R_b < BI/\Delta p$ either is not renegotiated (negative NPV, worse than not investing) or ends as an incentive-compatible contract that satisfies (IR$_l$) and could have been signed at the outset; so restricting to $R_b \geq BI/\Delta p$ loses nothing.

Intuition. A lender facing a borrower who is about to shirk is willing to forgive debt, because the higher stake raises the success probability and the lender keeps part of that gain. He forgives only if his remaining claim after forgiveness, the pledgeable part of the effort gain, beats what he gets

from a shirking borrower. Anticipating this, an initial contract that relies on forgiveness can at best reproduce the incentive-compatible contract: the multiplier k is the right one.

Common mistakes.

- Arguing that renegotiation always happens because restoring effort raises total surplus ($(\Delta p)RI > BI$). The lender gains only the *pledgeable* part of that surplus.
- Writing the lender's ex ante breakeven in (iii) as $p_H(RI - R_b) \geq I - A$, i.e. his initial claim with effort. He anticipates the renegotiation and values his final payoff, which with a take-it-or-leave-it offer by the borrower is $p_L(RI - R_b)$.
- Mixing the fixed- and variable-investment notation: here the private benefit is BI and the minimum stake is $BI/\Delta p$, not $B/\Delta p$.

Book answer. Matches the book's answer to (i)–(ii) (pp. 573–574), including the fictitious-project reading with cash $\hat{A} = p_L R_b$. The book gives no answer to (iii)–(iv); Steps 5–6 are this solution's.

Ref: Tirole §3.4.1–3.4.2, pp. 127–128; §3.2.2, pp. 116–117; answer pp. 573–574.

13. Loan size and collateral requirements — Tirole Exercise 4.17, p. 195

What is asked. The project has decreasing returns, $R(I)$ concave, and the entrepreneur can pledge costly collateral C that investors seize on failure. Assuming the first best cannot be financed, show that a poorer entrepreneur (lower A) or a worse agency problem (higher B, or higher $p_H/\Delta p$ with p_H fixed) leads to a smaller optimal investment and to more collateral.

Approach. This combines the variable-investment model (§3.4) with collateral pledging (§4.3.5); lecture Topic 4A, collateral reading. Collateral raises *pledgeable income* twice: investors get $\phi(C)$ on failure, and the borrower's stake grows, so her cash reward can shrink by C ($p_H C$). It costs a deadweight loss $C - \phi(C)$ on failure. We eliminate R_b with the binding (IC_b), write the Lagrangian with multiplier μ on (IR_l), derive the first-order conditions, and do comparative statics on the resulting system with the implicit-function theorem.

Notation.

- $I \geq 0$: investment; $R(I)$: income in success ($R(0) = 0$, $R' > 0$, $R'' < 0$, Inada).
- p_H, $p_L = p_H - \Delta p$; private benefit BI; $b \equiv p_H B/\Delta p$: per-unit agency cost (rises with B and, with p_H fixed, with $p_H/\Delta p$).
- $C \geq 0$: collateral pledged in failure, cost C to the entrepreneur, value $\phi(C)$ to investors ($\phi(0) = 0$, $\phi' > 0$, $\phi'' < 0$, $\phi'(0) = 1$, $\phi'(\infty) = 0$).
- $R_b \geq 0$: reward in success; A: cash; I^*: first-best size, $p_H R'(I^*) = 1$.
- $\mu \geq 0$: multiplier of the breakeven constraint; $\theta \equiv \mu/(1+\mu) \in [0, 1)$; $\varepsilon(I) \equiv -IR''(I)/R'(I) > 0$: curvature of R.

Assumption. The limited-liability constraint $R_b \geq 0$, i.e. $C \leq BI/\Delta p$, does not bind (as in footnote 24 of §4.3.5).

Step 1 — Programme. We set up the problem and eliminate R_b. The entrepreneur gets R_b on success and $-C$ on failure, so (IC_b) is $p_H R_b - (1 - p_H)C \geq p_L R_b - (1 - p_L)C + BI$, i.e. $(\Delta p)R_b + (\Delta p)C \geq BI$, i.e. $(\Delta p)(R_b + C) \geq BI$. Investors get $R(I) - R_b$ in success and $\phi(C)$ in failure. Collateral destroys $C - \phi(C) \geq 0$ in failure, so with binding breakeven her utility is the NPV:

$$\max_{I, C, R_b} U_b = p_H R(I) - I - (1 - p_H)[C - \phi(C)] \tag{78}$$

$$\text{s.t.} \quad (\Delta p)(R_b + C) \geq BI, \tag{IC_b}$$

$$p_H[R(I) - R_b] + (1 - p_H)\phi(C) \geq I - A. \tag{IR_l}$$

R_b does not enter (78) and lowers (IR_l), so (IC_b) binds: $R_b = BI/\Delta p - C$. Substituting into (IR_l),

$$p_H\Big[R(I) - \frac{BI}{\Delta p} + C\Big] + (1-p_H)\phi(C) \geq I - A$$
$$p_H R(I) - \frac{p_H B}{\Delta p}\, I + p_H C + (1-p_H)\phi(C) - I + A \geq 0$$
$$G(I,C) \equiv p_H R(I) - bI + p_H C + (1-p_H)\phi(C) - I + A \;\geq\; 0. \tag{79}$$

Collateral raises pledgeable income directly (ϕ) and by replacing reward ($p_H C$). The objective is concave and G is concave in (I, C), so the Kuhn–Tucker conditions are sufficient.

Step 2 — First-order conditions. We derive the conditions that characterise the optimum. With multiplier $\mu \geq 0$ on (79), the Lagrangian is

$$\mathcal{L}(I,C,\mu) = p_H R(I) - I - (1-p_H)\big[C - \phi(C)\big] + \mu\big[p_H R(I) - bI + p_H C + (1-p_H)\phi(C) - I + A\big].$$

Differentiating with respect to I and C and setting the derivatives to zero,

$$\frac{\partial \mathcal{L}}{\partial I} = p_H R'(I) - 1 + \mu\big[p_H R'(I) - b - 1\big] = 0,$$
$$\frac{\partial \mathcal{L}}{\partial C} = -(1-p_H)\big[1 - \phi'(C)\big] + \mu\big[p_H + (1-p_H)\phi'(C)\big] = 0.$$

The I-condition. Collect the terms in $p_H R'(I) - 1$:

$$(1+\mu)\big[p_H R'(I) - 1\big] - \mu b = 0$$
$$p_H R'(I) - 1 = \frac{\mu}{1+\mu}\, b.$$

The C-condition. Expand and collect the terms in $\phi'(C)$:

$$-(1-p_H) + (1-p_H)\phi'(C) + \mu p_H + \mu(1-p_H)\phi'(C) = 0$$
$$(1+\mu)(1-p_H)\phi'(C) = (1-p_H) - \mu p_H$$
$$\phi'(C) = \frac{1}{1+\mu} - \frac{\mu}{1+\mu}\,\frac{p_H}{1-p_H}.$$

Writing $\theta = \mu/(1+\mu)$, so that $1/(1+\mu) = 1-\theta$,

$$p_H R'(I) - 1 = \theta\, b, \tag{80}$$
$$\phi'(C) = 1 - \theta - \theta\frac{p_H}{1-p_H} = 1 - \theta\,\frac{(1-p_H)+p_H}{1-p_H} = 1 - \frac{\theta}{1-p_H}. \tag{81}$$

If $\mu = 0$, (80)–(81) give $I = I^*$, $C = 0$, which violates (79) by the assumption $p_H[R(I^*) - BI^*/\Delta p] < I^* - A$. So $\theta > 0$: (80) gives $I < I^*$ and (81) gives $\phi'(C) < 1 = \phi'(0)$, i.e. $C > 0$; finiteness of C needs $\theta < 1 - p_H$. These are the book's conditions, since $\theta = \mu/(1+\mu)$.

Step 3 — Monotonicity in θ. The FOCs define I and C as functions of θ (and b); we find how they move and then pin θ down with the binding breakeven. Totally differentiating (80), $p_H R''(I)\,dI = b\,d\theta + \theta\,db$; and (81), $\phi''(C)\,dC = -d\theta/(1-p_H)$. With $R'' < 0$, $\phi'' < 0$ and $Q \equiv 1/(p_H|R''(I)|) > 0$, dividing by $p_H R'' = -1/Q$ and by ϕ'':

$$\frac{\partial I}{\partial \theta} = -bQ < 0, \qquad \frac{\partial I}{\partial b} = -\theta Q < 0, \tag{82}$$
$$C'(\theta) = -\frac{1}{(1-p_H)\phi''(C)} > 0. \tag{83}$$

Let $g(\theta; A, b) \equiv G(I(\theta, b), C(\theta))$, the slack of (79) along the FOCs; the optimum is the root $g = 0$. Using (80), $p_H R' = 1 + \theta b$, so $G_I = p_H R' - b - 1 = (1 + \theta b) - b - 1 = -(1 - \theta)b < 0$; also $G_C = p_H + (1 - p_H)\phi' > 0$. Then, by the chain rule and (82)–(83),

$$\begin{aligned} g_\theta &= G_I \frac{\partial I}{\partial \theta} + G_C C'(\theta) \\ &= \big(-(1-\theta)b\big)\big(-bQ\big) + G_C C'(\theta) \\ &= (1-\theta)b^2 Q + G_C C'(\theta) \ > 0. \end{aligned} \tag{84}$$

g is < 0 at $\theta = 0$ (first best not pledgeable) and $\to +\infty$ as $\theta \to 1 - p_H$ ($C \to \infty$), so the root $\theta^* \in (0, 1 - p_H)$ is unique.

Step 4 — Comparative statics in A. $g_A = 1$, so by the implicit-function theorem and (84)

$$\frac{d\theta^*}{dA} = -\frac{g_A}{g_\theta} = -\frac{1}{g_\theta} < 0. \tag{85}$$

Combining with (82)–(83) by the chain rule:

$$\begin{aligned} \frac{dI}{dA} &= \frac{\partial I}{\partial \theta}\,\frac{d\theta^*}{dA} = (-bQ)\Big(-\frac{1}{g_\theta}\Big) = \frac{bQ}{g_\theta} > 0, \\ \frac{dC}{dA} &= C'(\theta)\,\frac{d\theta^*}{dA} = -\frac{C'(\theta)}{g_\theta} < 0. \end{aligned}$$

As A decreases, θ^* (hence $\mu = \theta/(1-\theta)$) rises, the investment size falls and the collateral rises, as claimed.

Answer (balance sheet). As A decreases, the shadow cost μ rises, investment I falls and collateral C rises: $dI/dA > 0$, $dC/dA < 0$.

Step 5 — Comparative statics in the agency cost b. Both B and $p_H/\Delta p$ (with p_H fixed) enter only through $b = p_H B/\Delta p$, so it suffices to vary b. Now b also enters $I(\theta, b)$; by (79), $G_b = -I$, so

$$\begin{aligned} g_b &= G_b + G_I \frac{\partial I}{\partial b} \\ &= -I + \big(-(1-\theta)b\big)\big(-\theta Q\big) = -I + \theta(1-\theta)bQ, \end{aligned} \tag{86}$$

and by the implicit-function theorem $d\theta^*/db = -g_b/g_\theta$. *Investment.* Using (82), (84) and (86) (with $D \equiv G_C C' > 0$),

$$\begin{aligned} \frac{dI}{db} &= \frac{\partial I}{\partial \theta}\frac{d\theta^*}{db} + \frac{\partial I}{\partial b} = (-bQ)\Big(-\frac{g_b}{g_\theta}\Big) + (-\theta Q) = -Q\Big(\theta - b\,\frac{g_b}{g_\theta}\Big) \\ &= -Q\,\frac{\theta g_\theta - b g_b}{g_\theta} = -Q\,\frac{\theta D + bI}{g_\theta} \ < 0, \end{aligned} \tag{87}$$

where the numerator is $\theta g_\theta - b g_b = \theta(1-\theta)b^2 Q + \theta D + bI - \theta(1-\theta)b^2 Q = \theta D + bI > 0$. So I falls with B and with $p_H/\Delta p$, as claimed. *Collateral.* $dC/db = C'(\theta)\,d\theta^*/db = -C'(\theta) g_b/g_\theta$ has the sign of $-g_b$, and $-g_b > 0$ iff $I > \theta(1-\theta)bQ$, i.e. (multiplying by $1/Q = p_H|R''|$) iff $p_H|R''|I > \theta(1-\theta)b$. By the definition of ε and (80), $p_H|R''|I = \varepsilon(I)\,p_H R'(I) = \varepsilon(I)(1 + \theta b)$, so

$$\frac{dC}{db} > 0 \iff \varepsilon(I)\,(1 + \theta b) \ > \ \theta(1-\theta)\,b. \tag{88}$$

This holds when R is concave enough (for instance $\varepsilon \geq 1$, since $\theta(1-\theta)b < 1 + \theta b$), but it can fail when R is close to linear. Counterexample: $R(I) = 3I^{0.95}$ ($\varepsilon = 0.05$), $\phi(C) = \ln(1 + C)$, $p_H = 0.5$, $\Delta p = 0.25$, $A = 1$. For $B = 0.5$ ($b = 1$) the solution is $\theta^* = 0.3276$, $I = 4.1203$, $C = 1.9003$; for $B = 2$ ($b = 4$) it is $\theta^* = 0.1238$, $I = 0.3815$, $C = 0.3292$. Collateral *falls* as the agency cost rises; at

$b = 1$, (88) reads $0.05(1 + 0.3276) = 0.0664 > 0.3276(1 - 0.3276)(1) = 0.2203$, false. (The first best is not pledgeable in both cases and $C \leq BI/\Delta p$.)

Answer (agency cost). Investment always falls with the agency cost, $dI/db < 0$. Collateral rises with it iff $\varepsilon(I)(1 + \theta b) > \theta(1 - \theta)b$ (e.g. whenever $\varepsilon \geq 1$); the statement's claim for C is not general, as the counterexample shows.

Intuition. A weaker balance sheet tightens the financing constraint; the entrepreneur pays for funding on both margins she controls, investing less (which saves outside funds) and pledging more collateral (which adds pledgeable income at a deadweight cost). A higher agency cost also lowers investment, but it has a second effect: each unit of investment now absorbs more pledgeable income, so the downsizing itself releases funds. When output is nearly linear in I, downsizing is cheap and strong enough that less collateral is needed. Only the balance-sheet comparative static is unambiguous.

Common mistakes.

- Dropping the indirect effect of collateral, the term $p_H C$: collateral also lets the cash reward R_b fall by C without breaking (IC$_b$).
- Treating b as a pure shift of the constraint. It also enters the I-condition (80), which is why the sign of dC/db is ambiguous.
- Ignoring limited liability, $R_b = BI/\Delta p - C \geq 0$: the substitution of collateral for reward is valid only while it holds.

Book answer. Agrees with the book's answer (p. 591): the FOCs (80)–(81) and "as the balance sheet deteriorates, μ increases, I decreases, and C increases". The book's answer does not claim that collateral rises with the agency cost (it says only that borrowing rises when the agency cost falls). The statement's claim for C holds only under (88); the claim for I holds in general (87).

Ref: Tirole §4.3.5, p. 170; §3.4.2, pp. 127–128; answer p. 591.

14. Dealing with asset substitution — Tirole Exercise 7.3, pp. 323–324

What is asked. (i) Without asset substitution: explain the threshold equation (1), give the entrepreneur's expected utility, and describe the optimal contracts (at least at $A = \bar{A}$). (ii) Now she can secretly run down maintenance (lower liquidation value L) to raise the success probability. Explain condition (2), find the smallest A for which this threat does no harm, and describe the contract that achieves it.

Approach. The fixed-investment model with a verifiable liquidation value (lecture Topic 4B, asset substitution and remedies). Liquidation value is not subject to moral hazard, so all of it is pledgeable; continuation income is pledgeable only up to $R - B/\Delta p$. A debt-like claim pays the entrepreneur only in continuation, so she prefers to shift value from liquidation to continuation: *asset substitution*. To prevent it she must be paid something in liquidation, conditional on good maintenance; that payment is a rent, which lowers *pledgeable income* and raises the minimum equity. We write (IR$_l$), (IC$_b$) and a no-substitution constraint, and minimise her rent.

Notation.

- $I > A$: fixed investment; A: cash; $x \in (0, 1)$: probability of demand; with probability $1 - x$ there is no demand and the assets are liquidated for L.
- R: income in success; p_H, $p_L = p_H - \Delta p$, B: as in Section 3.2.
- L_0: liquidation value without asset substitution; $\tau(L)$: change in success probability when maintenance yields liquidation value L ($\tau' < 0$, $\tau'' < 0$, $\tau(L_0) = 0$).
- r_b: entrepreneur's payment on liquidation; R_b: payment on continuation and success.

- U_b^*: entrepreneur's utility; $\bar{A}$: threshold (1); A^*: threshold with asset substitution.

Assumption. The statement's "(hence, with probability x)" for liquidation is read as probability $1-x$, the probability of no demand. $L \geq 0$ is chosen from $[0, \infty)$.

Step 1 — (i) Interpreting (1). The liquidation value L_0 is verifiable and not subject to moral hazard, so all of it can go to investors. On continuation only $R - R_b$ with $R_b \geq B/\Delta p$ can. The left-hand side of (1) is the maximal expected pledgeable income,

$$\mathcal{P}_0 = (1-x)L_0 + xp_H\Big(R - \frac{B}{\Delta p}\Big), \tag{89}$$

and (1) sets it equal to the investors' outlay at the threshold: financing is feasible iff $A \geq \bar{A} = I - \mathcal{P}_0$. The side condition $L_0 \leq p_H(R - B/\Delta p)$ says that, when there is demand, investors also prefer continuation (their maximal continuation claim beats L_0): there is no conflict over the continuation decision.

Answer (i), interpret (1). The left side of (1) is the maximal expected pledgeable income (all of L_0 on liquidation, $R - B/\Delta p$ on success after continuation); (1) defines $\bar{A}$ as the cash at which it just covers the investors' outlay $I - A$.

Step 2 — (i) Utility and optimal contracts. Competitive investors break even, so the entrepreneur receives the expected value created minus the investment, net of her own cash: $U_b^* = \big[(1-x)L_0 + xp_H R\big] - (I - A) - A$, i.e.

$$U_b^* = (1-x)L_0 + xp_H R - I. \tag{90}$$

A contract (r_b, R_b), liquidating iff there is no demand, is optimal iff

$$(1-x)(L_0 - r_b) + xp_H(R - R_b) = I - A, \tag{91}$$

$$R_b \geq \frac{B}{\Delta p}, \quad r_b \geq 0. \tag{92}$$

For $A > \bar{A}$ there is a continuum (the slack $A - \bar{A}$ can be paid as r_b or as extra R_b). At $A = \bar{A}$, subtracting (1), $I - \bar{A} = (1-x)L_0 + xp_H(R - B/\Delta p)$, from (91),

$$-(1-x)\, r_b - xp_H\Big(R_b - \frac{B}{\Delta p}\Big) = 0,$$

and both terms are ≤ 0 by (92), so each is zero: $r_b = 0$ and $R_b = B/\Delta p$. This is a debt contract with face value $D = R - B/\Delta p \geq L_0$ (since $L_0 \leq p_H(R - B/\Delta p) \leq D$). On liquidation the firm defaults and investors take L_0; on success they are repaid D.

Answer (i), utility and contracts. $U_b^* = (1-x)L_0 + xp_H R - I$. Optimal contracts: any (r_b, R_b) with $r_b \geq 0$, $R_b \geq B/\Delta p$ and binding (91); at $A = \bar{A}$ the unique one is debt with face $D = R - B/\Delta p$ ($r_b = 0$, $R_b = B/\Delta p$).

Step 3 — (ii) Interpreting (2). With maintenance level L (and effort), the NPV is

$$NPV(L) = (1-x)L + x\big[p_H + \tau(L)\big]R - I. \tag{93}$$

Differentiating, $NPV'(L) = (1-x) + x\tau'(L)R$ and $NPV''(L) = x\tau''(L)R < 0$, so it is strictly concave, and its first-order condition is $(1-x) + x\tau'(L)R = 0$, i.e. (dividing by x) $\tau'(L)R = -(1-x)/x$, which (2) says holds at L_0. So L_0 is the unique efficient maintenance level: one more unit of liquidation value, worth $1-x$ in expectation, costs exactly $x|\tau'|R$ of expected continuation income. $\tau(L_0) = 0$ normalises the success probabilities so that $NPV(L_0) = U_b^*$.

Answer (ii), interpret (2). (2) says L_0 is the efficient maintenance level: it maximises the NPV, and at L_0 the success probabilities are those of part (i).

Step 4 — (ii) The entrepreneur's maintenance choice. Under the step contract ($r_b(L) = r_b$ if $L \geq L_0$, 0 otherwise) and anticipating effort at stage 4 (her (IC$_b$), $(\Delta p)R_b \geq B$, does not depend on L), she chooses L to maximise

$$U(L) = (1-x)\, r_b(L) + x\big[p_H + \tau(L)\big]R_b. \tag{94}$$

For $L < L_0$ she gets 0 on liquidation and τ is decreasing, so the best deviation is $L = 0$; any $L > L_0$ lowers τ for no gain. So L_0 is chosen iff $U(L_0) \geq U(0)$:

$$\begin{aligned} (1-x)r_b + x\big[p_H + \tau(L_0)\big]R_b &\geq (1-x)(0) + x\big[p_H + \tau(0)\big]R_b \\ (1-x)r_b + xp_H R_b &\geq xp_H R_b + x\tau(0)R_b \qquad (\tau(L_0) = 0) \\ (1-x)\, r_b &\geq x\, \tau(0)\, R_b. \end{aligned} \tag{NS}$$

A debt contract ($r_b = 0$) violates (NS): the entrepreneur would run the assets down to gamble on continuation.

Step 5 — (ii) Minimum A. Financing at the efficient L_0 requires the investors' breakeven (91) with (IC$_b$) and (NS). The entrepreneur's expected rent $(1-x)r_b + xp_H R_b$ is minimised by (NS) at equality (any larger r_b is pure rent); substituting $(1-x)r_b = x\tau(0)R_b$, the rent is $x\tau(0)R_b + xp_H R_b = x\big[p_H + \tau(0)\big]R_b$, increasing in R_b, so $R_b = B/\Delta p$, the lowest value (IC$_b$) allows. Then

$$r_b = \frac{x\,\tau(0)}{1-x}\,\frac{B}{\Delta p}. \tag{95}$$

Substituting $R_b = B/\Delta p$ and (95) into (91) at equality,

$$\begin{aligned} I - A^* &= (1-x)(L_0 - r_b) + xp_H\Big(R - \frac{B}{\Delta p}\Big) \\ A^* &= \Big[I - (1-x)L_0 - xp_H\Big(R - \frac{B}{\Delta p}\Big)\Big] + (1-x)\, r_b \\ &= \bar{A} + (1-x)\, r_b = \bar{A} + x\,\tau(0)\,\frac{B}{\Delta p}, \end{aligned} \tag{96}$$

using (1) for the bracket and (95) for $(1-x)r_b$. For $A \geq A^*$, asset substitution is innocuous: the firm is financed, maintains L_0 and the entrepreneur gets U_b^*. For $A \in [\bar{A}, A^*)$ the efficient contract is not fundable. Illustration: $x = 0.6$, $p_H = 0.8$, $p_L = 0.4$, $B = 1$, $R = 10$, $L_0 = 3$, $I = 5.5$, $\tau(L) = (9 - L^2)/90$ (which satisfies (2): $\tau'(3)R = -(6/90)(10) = -2/3 = -(0.4/0.6)$): $U_b^* = 0.4(3) + 0.6(0.8)(10) - 5.5 = 1.2 + 4.8 - 5.5 = 0.5$, $B/\Delta p = 2.5$, $\bar{A} = 5.5 - 1.2 - 0.6(0.8)(10 - 2.5) = 5.5 - 1.2 - 3.6 = 0.7$, $\tau(0) = 9/90 = 0.1$, $r_b = (0.6)(0.1)(2.5)/(0.4) = 0.375$, $A^* = 0.7 + (0.4)(0.375) = 0.85$.

Step 6 — (ii) The optimal contract. It is debt plus a reward for good maintenance: the entrepreneur keeps $B/\Delta p$ on success and receives r_b on liquidation, but only if the assets are in good condition (a severance payment conditional on $L \geq L_0$). Giving her a stake in the liquidation value counters her bias toward the continuation state, where her debt-like claim pays.

Answer (ii), minimum A and contract. $A^* = \bar{A} + x\,\tau(0)\,B/\Delta p$. The contract is debt ($R_b = B/\Delta p$ on success) plus a payment $r_b = x\tau(0)B/[(1-x)\Delta p]$ on liquidation, made only if the assets were maintained at L_0.

Intuition. A claim that pays only in success makes the entrepreneur care only about continuation profit, so she cuts maintenance to raise the success probability: risk shifting at the expense of investors, who own the liquidation value. To stop it, she must share in the liquidation value. That share is an extra rent, so it lowers pledgeable income and raises the equity needed, by the expected gain from the most aggressive substitution, $x\,\tau(0)B/\Delta p$.

Common mistakes.

- Reading "liquidation … (hence, with probability x)" literally; liquidation happens when there is no demand, with probability $1-x$ (Figure 7.16).
- Keeping the debt contract of part (i) in part (ii). With $r_b = 0$ the no-substitution constraint fails and she runs the assets down to $L = 0$.
- Paying r_b on liquidation unconditionally. A payment that does not depend on L leaves her maintenance incentive unchanged; it must be conditional on $L \geq L_0$.

Book answer. Matches the book's answer (p. 602): the same interpretation of (1), U_b^*, the debt contract at $A = \bar{A}$, the step-function contract with $R_b = B/\Delta p$, the no-substitution inequality $(1-x)r_b + xp_H B/\Delta p \geq x[p_H + \tau(0)]B/\Delta p$ and the threshold $I - A^* = (1-x)(L_0 - r_b) + xp_H(R - B/\Delta p)$. The statement's "with probability x" for liquidation looks like a typo for $1-x$.

Ref: Tirole §3.2.2, pp. 116–117; §7.2.3 "Effort and Risk Taking", p. 314; answer p. 602.

15. A borrower-friendly bankruptcy court — Tirole Exercise 4.9, pp. 190–191

What is asked. (i) Argue that investors should get the short-term profit and the liquidation value, that the entrepreneur's continuation reward should be exactly $B/\Delta p$, and explain assumptions (1) and (2). (ii) Write the borrower's optimisation problem and the equation for the continuation cutoff r^*. (iii) Show that short-term debt $d = r^*$ implements the contract (default = liquidation) and how d depends on her cash A. (iv) A court splits the liquidation proceeds equally with the borrower; show that if $r^* > \hat{r}$ she loses access to financing. (v) If $r^* < \hat{r}$, show that the court either blocks financing or makes bankruptcy more likely, hurting the borrower but not lenders.

Approach. Lecture Topic 4B, bankruptcy design. The fixed-investment model with an intermediate verifiable profit r and a liquidation option. Everything not subject to moral hazard (r, L) is *pledgeable income* at no incentive cost; continuation income is pledgeable only up to $\rho_0(r)$. When the project cannot be financed with continuation in every state, the borrower buys funding by accepting inefficient liquidation where it raises pledgeable income most cheaply, which gives a cutoff rule. We write the Lagrangian, maximise it state by state, and use the investors' breakeven condition to find r^*; the court changes investors' payoff from liquidation from L to $L/2$.

Notation.

- $r \in [0, \bar{r}]$: verifiable short-term profit, density $f > 0$, c.d.f. F; $L > 0$: liquidation value.
- $p_H(r)$, $p_L(r) = p_H(r) - \Delta p$: success probabilities ($p_H' > 0$); R: final profit in success; B: private benefit.
- $\rho_1(r) = p_H(r)R$: continuation value; $\rho_0(r) = p_H(r)(R - B/\Delta p)$: pledgeable continuation value; both increasing in r.
- $\ell(r) \in \{0, 1\}$: liquidation policy; $R_b(r)$: reward on continuation and success; r^*: optimal cutoff; d: short-term debt.
- $\hat{r}$: $\rho_0(\hat{r}) = L/2$; r^{**}: cutoff under the borrower-friendly court.
- $P(c)$, $P_c(c)$: investors' expected income with cutoff c under an enforcing / borrower-friendly court.

Assumption. The court's equal split of L cannot be undone by contract.

Step 1 — (i) What goes to investors; $R_b = B/\Delta p$; conditions (1)–(2). r and L are not subject to moral hazard, so giving them to investors raises pledgeable income without hurting incentives. On continuation, (IC_b) is $[p_H(r) - p_L(r)]R_b \geq B$, i.e. $R_b \geq B/\Delta p$; any excess is wasted pledgeable income, which would force more liquidation (proved in Step 2). Condition (1), $\rho_1(r) > L$, says continuation is always efficient. Condition (2): always liquidating raises enough, $E[r] + L > I - A$,

but always continuing does not, $E[r+\rho_0(r)] < I-A$. Financing therefore needs inefficient liquidation in some states.

Answer (i). r and L go to investors because pledging them costs no incentives; $R_b = B/\Delta p$ because any larger reward wastes pledgeable income. (1): continuation is always efficient. (2): liquidating always would raise enough funds, continuing always would not, so some inefficient liquidation is the price of financing.

Step 2 — (ii) Programme and the cutoff. With binding breakeven the borrower's utility is the NPV:

$$\max_{\ell(\cdot),R_b(\cdot)} \; E[r] + \int_0^{\bar r} \big[\ell(r)L + (1-\ell(r))\rho_1(r)\big] f(r)\,dr - I \tag{97}$$

$$\text{s.t.} \quad E[r] + \int_0^{\bar r} \big[\ell(r)L + (1-\ell(r))p_H(r)(R-R_b(r))\big] f(r)\,dr \;\geq\; I-A, \tag{98}$$

$$R_b(r) \geq B/\Delta p. \tag{99}$$

Let μ be the multiplier of (98). $\mu > 0$: the unconstrained optimum, $\ell \equiv 0$ by (1), violates (98) by (2). R_b does not enter (97) and lowers (98), so $R_b(r) = B/\Delta p$ and the investors' continuation income is $p_H(r)(R - B/\Delta p) = \rho_0(r)$. The Lagrangian is then

$$\mathcal{L} = (1+\mu)E[r] - I - \mu(I-A) + \int_0^{\bar r} \Big\{\ell(r)\,(1+\mu)L + (1-\ell(r))\,\big[\rho_1(r) + \mu\rho_0(r)\big]\Big\} f(r)\,dr,$$

which is linear in $\ell(r)$ state by state. Maximising the integrand pointwise: liquidate iff $(1+\mu)L > \rho_1(r) + \mu\rho_0(r)$, i.e.

$$\mu\big[L - \rho_0(r)\big] \;>\; \rho_1(r) - L. \tag{100}$$

The left-hand side decreases and the right-hand side increases in r, so the policy is a cutoff: liquidate iff $r < r^*$. At the cutoff (100) holds with equality and $\rho_1(r^*) > L$, so $\mu[L - \rho_0(r^*)] = \rho_1(r^*) - L > 0$ and, with $\mu > 0$, $\rho_0(r^*) < L$.

Step 3 — (ii) The equation for r^*. With cutoff c, investors receive

$$P(c) = E[r] + L\,F(c) + \int_c^{\bar r} \rho_0(r) f(r)\,dr, \tag{101}$$

and the NPV, $E[r] + LF(c) + \int_c^{\bar r} \rho_1 f\,dr - I$, falls with c by (1). So r^* is the smallest root of

$$E[r] + L\,F(r^*) + \int_{r^*}^{\bar r} \rho_0(r) f(r)\,dr = I - A. \tag{102}$$

By (2), $P(0) = E[r+\rho_0(r)] < I - A < E[r] + L = P(\bar r)$, so by continuity $r^* \in (0, \bar r)$ (the statement's "$(0, r)$" should read $(0, \bar r)$).

Answer (ii). $\max_{\ell(\cdot),R_b(\cdot)}$ (97) s.t. (98)–(99); continue iff $r \geq r^*$, with r^* the smallest root of $E[r] + LF(r^*) + \int_{r^*}^{\bar r} \rho_0(r) f(r)\,dr = I - A$.

Step 4 — (iii) Short-term debt and the effect of A. Implementation: short-term debt $d = r^*$ due at date 1, any excess $r - d$ also paid to investors (for example as a payout to outside shareholders), and a long-term claim of $R - B/\Delta p$ on success. If $r \geq d$ the debt is paid and the firm continues; if $r < d$ it defaults, creditors take control and liquidate ("bankruptcy"). They do not renegotiate: for $r < r^*$, $\rho_0(r) < \rho_0(r^*) < L$, and the borrower has no cash to bribe them. By Leibniz's rule applied to (101),

$$\frac{dP}{dc} = L\,f(c) - \rho_0(c) f(c) = \big[L - \rho_0(c)\big] f(c).$$

Differentiating (102), $P(r^*) = I - A$, with respect to A:

$$[L - \rho_0(r^*)]f(r^*)\,dr^* = -dA$$
$$\frac{dr^*}{dA} = -\frac{1}{[L - \rho_0(r^*)]f(r^*)} \; < 0. \tag{103}$$

Short-term debt falls with initial equity: a weaker balance sheet needs more pledgeable income, so the firm must accept bankruptcy in more states.

Answer (iii). Short-term debt $d = r^*$ (default and liquidation iff $r < d$), plus a long-term claim $R - B/\Delta p$ on success, implements the optimum; $dd/dA = dr^*/dA = -1/\{[L - \rho_0(r^*)]f(r^*)\} < 0$.

Step 5 — (iv) Borrower-friendly court with $r^* > \hat{r}$. Liquidation now yields investors only $L/2$. For any policy, investors' income in state r (besides r) is at most $\max\{L/2, \rho_0(r)\}$. Compare with the enforcing-court contract state by state. For $r < r^*$, $\max\{L/2, \rho_0(r)\} < L$, since $L/2 < L$ and $\rho_0(r) < \rho_0(r^*) < L$. For $r \geq r^* > \hat{r}$, $\rho_0(r) > \rho_0(\hat{r}) = L/2$, so the maximum is $\rho_0(r)$, the same as before. Hence

$$E[r] + \int_0^{\bar{r}} \max\Big\{\frac{L}{2}, \rho_0(r)\Big\} f(r)\,dr \;<\; E[r] + LF(r^*) + \int_{r^*}^{\bar{r}} \rho_0 f\,dr = I - A, \tag{104}$$

using (102). No policy lets investors break even: the borrower loses access to financing.

Answer (iv). With $r^* > \hat{r}$, the investors' maximal expected income under the court is below $I - A$: no contract is financed.

Step 6 — (v) Court with $r^* < \hat{r}$. Cutoff policies are again optimal (same argument as (100) with $L/2$ for investors). Investors' income with cutoff c is

$$P_c(c) = E[r] + \frac{L}{2}F(c) + \int_c^{\bar{r}} \rho_0(r) f(r)\,dr, \qquad P_c'(c) = \Big[\frac{L}{2} - \rho_0(c)\Big] f(c), \tag{105}$$

which rises up to $\hat{r}$ and falls after it, so its maximum is $P_c(\hat{r})$. If $P_c(\hat{r}) < I - A$, there is no financing. Otherwise the borrower picks the smallest root r^{**} of $P_c(c) = I - A$. For $0 < c \leq r^*$, $P_c(c) < P(c) \leq I - A$ by (101) and (102), so $r^{**} \in (r^*, \hat{r}]$: bankruptcy is more likely. Lenders break even in either regime, so they are not hurt. The borrower gets the NPV, which falls with the cutoff (Step 3), so $NPV(r^{**}) < NPV(r^*)$: she is hurt. More directly, with an enforcing court she could have written the $L/2$ split into her contract; the friendly court only removes options.

Answer (v). If $P_c(\hat{r}) < I - A$ there is no financing; otherwise the cutoff rises to $r^{**} \in (r^*, \hat{r}]$ (more bankruptcy). Lenders still break even; the borrower's utility, the NPV, falls.

Step 7 — Numerical illustration. $r \sim U[0,1]$, $p_H(r) = 0.5 + 0.4r$, $\Delta p = 0.2$, $R = 10$, $B = 1.2$ ($R - B/\Delta p = 4$), $L = 4.5$. Then $\rho_1(r) = 5 + 4r > L$, $\rho_0(r) = 2 + 1.6r$, $E[r] = 0.5$, $P(0) = E[r + \rho_0] = 3.3$, $P(1) = E[r] + L = 5$, so (2) holds for $I - A \in (3.3, 5)$; $\hat{r} = (2.25 - 2)/1.6 = 0.15625$, and the court's maximal income is $P_c(\hat{r}) = 3.319531$. With $f = 1$ on $[0,1]$, (101) and (105) integrate to

$$P(c) = 0.5 + 4.5c + 2(1 - c) + 0.8(1 - c^2) = 3.3 + 2.5c - 0.8c^2,$$
$$P_c(c) = 0.5 + 2.25c + 2(1 - c) + 0.8(1 - c^2) = 3.3 + 0.25c - 0.8c^2,$$

so $P_c(\hat{r}) = 3.3 + 0.25(0.15625) - 0.8(0.15625)^2 = 3.3 + 0.0390625 - 0.0195313 = 3.319531$. (iv) $I - A = 4$: $P(r^*) = 4$ gives $0.8r^{*2} - 2.5r^* + 0.7 = 0$, smallest root $r^* = (2.5 - \sqrt{6.25 - 2.24})/1.6 = 0.3109 > \hat{r}$; $\rho_0(r^*) = 2 + 1.6(0.3109) = 2.4975 < L$; and $3.319531 < 4$: no financing. (v) $I - A = 3.31$: $2.5r^* - 0.8r^{*2} = 0.01$ gives $r^* = 0.0040 < \hat{r}$; under the court $0.25r^{**} - 0.8r^{**2} = 0.01$ gives $r^{**} = (0.25 - \sqrt{0.0625 - 0.032})/1.6 = 0.0471$, and the NPV loss is $\int_{r^*}^{r^{**}} (\rho_1 - L)\,dr = \int_{r^*}^{r^{**}} (0.5 + 4r)\,dr = 0.0260$.

Intuition. Liquidation is the investors' collateral: it is what they get when the firm cannot pay. A court that hands half of it to the borrower lowers the value of default to creditors. To keep

them whole the contract must liquidate more often, which destroys value, or financing breaks down. Lenders still break even, so the whole cost of the borrower-friendly rule falls on the borrower it was meant to protect.

Common mistakes.

- Liquidating whenever $\rho_0(r) < L$. Liquidation is always inefficient here (1); it is used only as far as needed to break even, so the cutoff comes from the binding breakeven, not from $\rho_0 = L$.
- Concluding that a court that hands the borrower half of L helps her. Lenders must still break even, so the borrower pays for the transfer through more liquidation or no financing.
- Signing dr^*/dA wrong: more cash means less pledgeable income is needed, so fewer liquidations and *less* short-term debt.

Book answer. Matches the book's answer (pp. 586–587): the reasons for pledging r and L, $R_b = B/\Delta p$, the programme and r^* as the lowest root of breakeven with $L \geq \rho_0(r^*)$, the debt implementation $d = r^*$, $dr^*/dA < 0$, and (iv)–(v) as read from the book's Figures 2–3 (bankruptcy on $[0, r^{**}]$ with $r^* < r^{**} \leq \hat{r}$, lenders not hurt, borrower hurt). The statement's "$(0, r)$" is a typo for $(0, \bar{r})$.

Ref: Tirole §3.2.2, pp. 116–117; Exercise 4.9, pp. 190–191; answer pp. 586–587.

16. Default as insurance: the default cutoff — Original exercise

Data (full statement: list, exercise 16; built on Topic 4B slides 41–44, Indarte 2023). Debt payment $D = 12$; unseizable liquid assets $A = 5$; income $Y \sim U[5, 25]$, so $F(a) = (a - 5)/20$; utility $u(C) = \sqrt{C}$; in default the household keeps $E = 4$, creditors take $Y - E$ up to D, and the household bears a utility cost $\sigma = 1$.

What is asked. (a) Find the incomes at which the household simply does not have the cash to pay, and their probability. (b) Among the incomes at which it could pay, find where it prefers not to: derive the cutoff Y^* and check that it sits between $D - A$ and $D + E$. (c) Turn the cutoff into a default probability and split it into "cannot pay" and "will not pay". (d) Apply the rule to three concrete incomes. (e) Find what creditors expect to collect when default happens, and say why default is a form of insurance.

Approach. This is the lecture's *default as insurance* model (slides 41–44). In every income state the household compares two numbers: the utility of repaying, $u(A + Y - D)$, and the utility of defaulting, $u(A + E) - \sigma$. The first rises with Y and the second does not depend on Y, so there is a single income Y^* below which default wins. The default probability is then just $F(Y^*)$.

Notation.

- $D = 12$: debt payment due; $A = 5$: liquid assets creditors cannot seize; $Y \in [5, 25]$: income, cdf F.
- $E = 4$: income the court lets the household keep in default; $Y - E$: garnished income (capped at D).
- $\sigma = 1$: non-pecuniary cost of default, in utility units; $u(C) = \sqrt{C}$: utility of consumption $C \geq 0$.
- $c_r \equiv A + Y - D$: consumption if the household repays; $c_d \equiv A + E = 9$: consumption if it defaults.
- Y^*: default cutoff; $p = F(Y^*)$: default probability.

Step 1 — (a) Cash-flow default. The household can pay only if its liquid assets plus income

cover the debt, so it is forced into default when

$$\begin{aligned} A + Y &< D \\ Y &< D - A \\ Y &< 12 - 5 = 7. \end{aligned} \tag{106}$$

With the uniform cdf,

$$\Pr[Y < 7] = F(7) = \frac{7-5}{20} = \frac{2}{20} = 0.1. \tag{107}$$

Answer (a). Cash-flow default for $Y < 7$, probability 0.1.

Step 2 — (b) The strategic-default condition and the cutoff. For $Y \geq 7$ the household can pay and picks the option with higher utility. Repaying gives

$$u(A + Y - D) = \sqrt{5 + Y - 12} = \sqrt{Y - 7}, \tag{108}$$

and defaulting gives

$$u(A + E) - \sigma = \sqrt{5 + 4} - 1 = \sqrt{9} - 1 = 3 - 1 = 2 > 0. \tag{109}$$

So it defaults strategically when $\sqrt{Y - 7} < 2$. The cutoff equates the two:

$$\begin{aligned} \sqrt{Y^* - 7} &= 2 \\ Y^* - 7 &= 2^2 = 4 \\ Y^* &= 11. \end{aligned} \tag{110}$$

Bounds: $D - A = 7 < 11$, and $D + E = 12 + 4 = 16 > 11$. The first says that some households who *can* pay do not. The second matters for creditors: in every default state $Y < 11$, so the garnishment is $Y - E < 11 - 4 = 7 < 12 = D$; the cap at D never binds, creditors take all of $Y - E$, and the household keeps exactly $A + E$, as the slides assume. (If $Y - E$ exceeded D, "default" would leave the household with $A + Y - D$, the same as repaying, minus σ, and it would never choose it.)

Answer (b). Strategic default iff $7 \leq Y$ and $\sqrt{Y - 7} < 2$; $u(A + E) - \sigma = 2 > 0$; $Y^* = 11$, with $7 < 11 < 16$.

Step 3 — (c) Default probability. The household defaults for every $Y < Y^*$, whatever the reason:

$$p = F(Y^*) = F(11) = \frac{11-5}{20} = \frac{6}{20} = 0.3. \tag{111}$$

Of this, $F(7) = 0.1$ is cash-flow default (107) and the rest is strategic:

$$F(11) - F(7) = 0.3 - 0.1 = 0.2. \tag{112}$$

Answer (c). $p = 0.3$: 0.1 cash-flow default and 0.2 strategic default.

Step 4 — (d) Three incomes. We apply (106) and (110).

- $Y = 6$: $A + Y = 5 + 6 = 11 < 12$, so it cannot pay (cash-flow default). It consumes $A + E = 9$, utility $\sqrt{9} - 1 = 2$; creditors receive $Y - E = 6 - 4 = 2$.
- $Y = 9$: $A + Y = 14 \geq 12$, so it can pay. Repaying gives $C = 5 + 9 - 12 = 2$ and $u = \sqrt{2} = 1.414214 < 2$, so it defaults strategically: $C = 9$, utility 2; creditors receive $9 - 4 = 5$.
- $Y = 20$: repaying gives $C = 5 + 20 - 12 = 13$ and $u = \sqrt{13} = 3.605551 > 2$, so it repays; creditors receive $D = 12$.

Answer (d). $Y = 6$: cash-flow default ($C = 9$, $u = 2$, creditors 2). $Y = 9$: strategic default ($C = 9$, $u = 2$, creditors 5). $Y = 20$: repays ($C = 13$, $u = 3.605551$, creditors 12).

Step 5 — (e) Recovery in default and the consumption floor. Conditional on default, Y is uniform on $[5, 11)$, so

$$E[Y \mid Y < 11] = \frac{5 + 11}{2} = 8,$$
$$E[Y - E \mid Y < 11] = 8 - 4 = 4. \tag{113}$$

The household's consumption never falls below $A + E = 9$: in all states $Y < 11$ it consumes 9.

Answer (e). Creditors expect to collect 4 per defaulted loan (out of 12 due). The consumption floor is $A + E = 9$. Default insures: it swaps the low and risky consumption $A + Y - D$ of bad income states for a fixed $A + E$, paid for by the creditors' losses (and by σ); a risk-averse household values that.

Intuition. The household holds a put option on its debt. Exercising it costs σ and yields the consumption floor $A + E$, so it is exercised exactly when repayment would leave consumption below the level at which the floor, net of σ, is better. Two thirds of the defaults (0.2 of 0.3) are strategic: these households could pay but the insurance is worth more to them than the stigma.

Common mistakes.

- Calling all default "inability to pay". Here only $Y < 7$ is cash-flow default; on $[7, 11)$ default is a choice.
- Comparing consumption instead of utility: σ is a utility cost, so the comparison is $\sqrt{Y - 7}$ versus $3 - 1$, not $Y - 7$ versus 9.
- Giving creditors D or 0 in default. They receive the garnishment $Y - E$, which is below D in every default state here.

Ref: Prof. Ricca, Topic 4B slides 41–44 (Aula/Topic4_B.pdf); Indarte (2023), *Journal of Finance* 78(5). No book exercise exists.

17. Default as insurance: comparative statics, the price of credit and ex ante welfare — Original exercise

Data (full statement: list, exercise 17; built on Topic 4B slides 37–44). $A = 5$, $E = 4$, $Y \sim U[5, 25]$ ($F(a) = (a - 5)/20$, $F' = 1/20$), $u(C) = \sqrt{C}$, default cost σ in utility; baseline $D = 12$, $\sigma = 1$, $Y^* = 11$, $p = 0.3$. Competitive risk-neutral lenders, riskless rate 0. The household must raise $L = 9.6$ in parts (c)–(e).

What is asked. (a) Find how the cutoff and the default probability move with the debt, the protected assets, the stigma cost and the exemption, first in general and then at the baseline. (b) Price the baseline claim: what is it worth to a lender, and what promised rate does that imply? (c) Compare, from the household's point of view before income is known, a world where debt must always be repaid with the baseline bankruptcy system, when in both the lender is paid fairly for lending 9.6. (d) Redo the bankruptcy side with a much smaller stigma cost. (e) Explain why the ranking flips, and find the stigma cost at which it flips.

Approach. Part (a) is the implicit-function theorem applied to the cutoff equation of slide 43; the sign of each derivative comes from concavity of u. Parts (b)–(e) add the lenders' side: a competitive lender prices *anticipated* default, so the household pays for its insurance through a higher face value. Because lenders break even and σ is a utility (not a resource) cost, the household's expected consumption is the same with and without bankruptcy. Bankruptcy then only reshapes consumption risk (a gain, by risk aversion) at the price of the deadweight σ in default states. Welfare depends on which is larger. This is the ex ante side of soft debt on slides 37–40.

Notation.

- $c_r \equiv A + Y^* - D$: consumption at the cutoff if the household repays; $c_d \equiv A + E$: consumption in default.
- $G(Y^*, x) \equiv u(A + Y^* - D) - u(A + E) + \sigma = 0$: the cutoff equation; $x \in \{D, A, \sigma, E\}$.
- $Q(D)$: lender's value of a claim with face D (expected repayment); $D/Q - 1$: promised rate.
- Regime N: no default, debt D_N; Regime B: bankruptcy, debt D_B with $Q(D_B) = L = 9.6$.
- EU: expected utility net of σ in default states; $CE = u^{-1}(EU) = EU^2$: certainty equivalent.

Step 1 — A sign lemma. All signs in (a) rest on the household consuming less at the cutoff when it repays than when it defaults. From $G = 0$ and $\sigma > 0$,

$$u(c_r) = u(c_d) - \sigma < u(c_d) \implies c_r < c_d \implies u'(c_r) > u'(c_d), \tag{114}$$

the first implication because u is increasing, the second because u is strictly concave.

Step 2 — (a) Implicit differentiation. The partial derivatives of G are

$$G_{Y^*} = u'(c_r), \quad G_D = -u'(c_r), \quad G_A = u'(c_r) - u'(c_d), \quad G_\sigma = 1, \quad G_E = -u'(c_d). \tag{115}$$

By the implicit-function theorem, $dY^*/dx = -G_x/G_{Y^*}$, so

$$\frac{dY^*}{dD} = -\frac{-u'(c_r)}{u'(c_r)} = 1 > 0, \tag{116}$$

$$\frac{dY^*}{dA} = -\frac{u'(c_r) - u'(c_d)}{u'(c_r)} = \frac{u'(c_d)}{u'(c_r)} - 1 < 0, \tag{117}$$

$$\frac{dY^*}{d\sigma} = -\frac{1}{u'(c_r)} < 0, \tag{118}$$

$$\frac{dY^*}{dE} = -\frac{-u'(c_d)}{u'(c_r)} = \frac{u'(c_d)}{u'(c_r)} > 0, \tag{119}$$

where the sign of (117) uses $u'(c_d)/u'(c_r) < 1$ from (114).

Step 3 — Baseline values. At the baseline $c_r = A + Y^* - D = 5 + 11 - 12 = 4$ and $c_d = 5 + 4 = 9$. With $u'(C) = 1/(2\sqrt{C})$:

$$u'(4) = \frac{1}{2 \cdot 2} = 0.25, \qquad u'(9) = \frac{1}{2 \cdot 3} = \frac{1}{6}, \qquad \frac{u'(9)}{u'(4)} = \frac{1/6}{1/4} = \frac{2}{3}. \tag{120}$$

Substituting in (116)–(119):

$$\frac{dY^*}{dD} = 1, \quad \frac{dY^*}{dA} = \frac{2}{3} - 1 = -\frac{1}{3}, \quad \frac{dY^*}{d\sigma} = -\frac{1}{0.25} = -4, \quad \frac{dY^*}{dE} = \frac{2}{3}.$$

Multiplying by $F'(Y^*) = 1/20$ gives the effects on p:

$$\frac{dp}{dD} = \frac{1}{20} = 0.05, \quad \frac{dp}{dA} = -\frac{1}{3 \cdot 20} = -\frac{1}{60}, \quad \frac{dp}{d\sigma} = -\frac{4}{20} = -0.2, \quad \frac{dp}{dE} = \frac{2}{3 \cdot 20} = \frac{1}{30}. \tag{121}$$

Slide 44's formula is $\partial p/\partial E = F'(Y^*)\, u'(A + E)/u'(A + Y^* - D)$, which is $F'(Y^*)$ times (119): $\frac{1}{20} \cdot \frac{2}{3} = \frac{1}{30}$, the same.

Why more A lowers default: an extra unit of protected assets raises consumption by one unit in *both* options. The utility gain is $u'(c_r)$ if the household repays and $u'(c_d)$ if it defaults, and $u'(c_r) > u'(c_d)$ because repaying leaves it poorer at the margin. Repaying gains more, so the household at the old cutoff now strictly prefers to repay, and Y^* falls. A richer household needs the insurance less.

Answer (a). $dY^*/dD = 1$, $dY^*/dA = u'(c_d)/u'(c_r) - 1 < 0$, $dY^*/d\sigma = -1/u'(c_r) < 0$, $dY^*/dE = u'(c_d)/u'(c_r) > 0$. Baseline: 1, $-1/3$, -4, $2/3$; $dp/dD = 0.05$, $dp/dA = -1/60$, $dp/d\sigma = -0.2$, $dp/dE = 1/30$, matching slide 44.

Step 4 — (b) The value of the baseline claim. The lender receives D when $Y \geq Y^*$ and $Y - E$ when $Y < Y^*$, so

$$\begin{aligned} Q(D) &= [1 - F(Y^*)]D + \frac{1}{20}\int_5^{Y^*} (y - E)\, dy \\ &= (1 - 0.3)(12) + \frac{1}{20}\left[\frac{(y-4)^2}{2}\right]_5^{11} \\ &= 8.4 + \frac{1}{20}\cdot\frac{7^2 - 1^2}{2} = 8.4 + \frac{48}{40} = 8.4 + 1.2 = 9.6. \end{aligned} \tag{122}$$

A borrower who receives 9.6 and promises 12 pays

$$\frac{D}{Q} - 1 = \frac{12}{9.6} - 1 = 1.25 - 1 = 0.25. \tag{123}$$

Answer (b). $Q = 9.6$; promised rate 25%, all of it a default premium (the riskless rate is 0).

Step 5 — (c) Regime N. With full enforcement the debt is riskless, so $D_N = L = 9.6$ and the household consumes $C_N = A + Y - 9.6 = Y - 4.6 \in [0.4, 20.4]$. Its expected utility is

$$\begin{aligned} EU_N &= \frac{1}{20}\int_5^{25} \sqrt{y - 4.6}\, dy = \frac{1}{20}\cdot\frac{2}{3}\Big[(y-4.6)^{3/2}\Big]_5^{25} \\ &= \frac{1}{30}\left[(20.4)^{3/2} - (0.4)^{3/2}\right] = \frac{92.139373 - 0.252982}{30} = \frac{91.886391}{30} = 3.062880, \end{aligned} \tag{124}$$

so $CE_N = 3.062880^2 = 9.381232$. Expected consumption: $E[C_N] = A + E[Y] - 9.6 = 5 + 15 - 9.6 = 10.4$.

Step 6 — (c) Regime B with $\sigma = 1$. By (b), $Q(12) = 9.6$, and Q increases in D below $D = 21$ (Step 7), so $D_B = 12$ and $Y^* = 11$, $p = 0.3$. In default ($Y < 11$) utility is $u(9) - 1 = 2$; when repaying, $C = Y - 7$:

$$\begin{aligned} EU_B &= p\,[u(A + E) - \sigma] + \frac{1}{20}\int_{11}^{25} \sqrt{y - 7}\, dy \\ &= 0.3(2) + \frac{1}{30}\left[18^{3/2} - 4^{3/2}\right] = 0.6 + \frac{76.367532 - 8}{30} = 0.6 + 2.278918 = 2.878918, \end{aligned} \tag{125}$$

so $CE_B = 2.878918^2 = 8.288168$. Expected consumption:

$$E[C_B] = 0.3(9) + \frac{1}{20}\int_{11}^{25} (y - 7)\, dy = 2.7 + \frac{1}{20}\cdot\frac{18^2 - 4^2}{2} = 2.7 + \frac{308}{40} = 2.7 + 7.7 = 10.4.$$

Answer (c). $D_N = 9.6$ (rate 0), $EU_N = 3.062880$, $CE_N = 9.381232$; $D_B = 12$ (rate 25%), $EU_B = 2.878918$, $CE_B = 8.288168$; $E[C] = 10.4$ in both. The household prefers Regime N.

Step 7 — (d) The general pricing formula. From $\sqrt{A + Y^* - D} = \sqrt{A + E} - \sigma$, squaring,

$$A + Y^* - D = \left(\sqrt{A + E} - \sigma\right)^2 \equiv c_r \quad \Longrightarrow \quad Y^* = D - A + c_r. \tag{126}$$

Write $Y^* = D + k$ with $k \equiv c_r - A = c_r - 5$. Multiplying (122) by 40 and evaluating the integral as in Step 4,

$$\begin{aligned} 40\,Q &= 2(25 - Y^*)D + (Y^* - 4)^2 - 1 \\ &= 2(25 - D - k)D + (D + k - 4)^2 - 1 \\ &= 50D - 2D^2 - 2kD + D^2 + 2(k - 4)D + (k - 4)^2 - 1 \\ &= -D^2 + 42D + (k - 4)^2 - 1, \end{aligned} \tag{127}$$

and $k-4=c_r-9$. (Check at $\sigma=1$: $c_r=4$, $-144+504+25-1=384=40(9.6)$.) The slope $42-2D$ is positive for $D<21$, so the lender accepts the smallest root. With $\sigma=0.1$: $c_r=(3-0.1)^2=8.41$ and $(c_r-9)^2=0.3481$. Setting $40Q=384$,

$$\begin{aligned} -D^2+42D+0.3481-1&=384\\ D^2-42D+384.6519&=0\\ D_B=21-\sqrt{441-384.6519}&=21-\sqrt{56.3481}=21-7.506537=13.493463. \end{aligned} \tag{128}$$

Then, from (126),

$$\begin{aligned} Y^*&=13.493463-5+8.41=16.903463,\\ p&=\frac{16.903463-5}{20}=0.595173,\\ \frac{D_B}{9.6}-1&=1.405569-1=0.405569. \end{aligned}$$

The formula applies: $5\le Y^*\le 25$, and the garnishment cap does not bind since $Y^*-E=12.903463<D_B$ (in general $D_B-(Y^*-E)=9-c_r>0$).

Step 8 — (d) Expected utility with $\sigma=0.1$. In default utility is $u(9)-0.1=2.9$; when repaying $C=5+y-D_B$, which equals $c_r=8.41$ at $y=Y^*$ and $30-D_B=16.506537$ at $y=25$:

$$\begin{aligned} EU_B&=p(2.9)+\frac{1}{30}\Big[(30-D_B)^{3/2}-c_r^{3/2}\Big]\\ &=0.595173(2.9)+\frac{16.506537^{3/2}-8.41^{3/2}}{30}\\ &=1.726002+\frac{67.063152-24.389}{30}=1.726002+1.422472=3.148474, \end{aligned} \tag{129}$$

so $CE_B=3.148474^2=9.912888$, above $CE_N=9.381232$. Expected consumption is again 10.4 (Step 9).

Answer (d). $D_B=13.493463$, $Y^*=16.903463$, $p=0.595173$, promised rate 40.56%; $EU_B=3.148474$, $CE_B=9.912888>9.381232$: now the household prefers bankruptcy (Regime B).

Step 9 — (e) Why the ranking flips. First, expected consumption is the same in both regimes because the lender breaks even and σ costs no resources:

$$E[C_B]=A+E[Y]-E[\text{payment to lender}]=5+15-9.6=10.4=E[C_N].$$

Second, C_B is less risky than C_N. Their difference is

$$C_B-C_N=\begin{cases}(A+E)-(A+Y-9.6)=13.6-Y, & Y<Y^*,\\ (A+Y-D_B)-(A+Y-9.6)=9.6-D_B<0, & Y\ge Y^*,\end{cases}$$

which falls with Y and changes sign once, from positive to negative. Two consumption plans with the same mean and a single crossing are ranked by every risk-averse household: the one higher in bad states wins, so $E[u(C_B)]>E[u(C_N)]$. Write the gap as an insurance gain minus a deadweight cost:

$$EU_B-EU_N=\underbrace{E[u(C_B)]-EU_N}_{\text{insurance gain}>0}-\underbrace{\sigma p}_{\text{stigma cost}}. \tag{130}$$

With $\sigma=1$: $E[u(C_B)]=2.878918+0.3=3.178918$, gain $=3.178918-3.062880=0.116038<\sigma p=0.3$. With $\sigma=0.1$: $E[u(C_B)]=3.148474+0.059517=3.207991$, gain $=0.145111>\sigma p=0.059517$. Bisection on $EU_B(\sigma)=EU_N$, recomputing $D_B(\sigma)=21-\sqrt{56+(c_r-9)^2}$ each time, gives $\bar\sigma=0.264820$.

Answer (e). Lenders price anticipated default, so bankruptcy is actuarially fair insurance in consumption terms: same mean (10.4), less risk. It raises welfare when the insurance gain exceeds the expected stigma cost σp: yes for $\sigma = 0.1$ ($0.145 > 0.060$), no for $\sigma = 1$ ($0.116 < 0.3$). Indifference at $\bar{\sigma} \approx 0.26$.

Intuition. Default is a state-contingent repayment: households pay a lot when income is high and little when it is low. Lenders anticipate this and charge for it (25% or 40.6% instead of 0%), so the household pays the full cost of the insurance. What it gets is a smoother consumption profile, which a risk-averse household values. What society loses is only the stigma σ each time default happens. A lower σ makes default more frequent (p doubles) and credit dearer, yet welfare rises: more default is not bad in itself. This model has no effort choice; with moral hazard at the borrowing stage (slides 37–40), softer default would also weaken incentives and add a further ex ante cost.

Common mistakes.

- Comparing the regimes at the same face value D. At a fixed D the default option can only help the household; the real comparison re-prices the debt so the lender breaks even.
- Reading a higher default rate or a higher interest rate as lower welfare. Both rise in (d) while welfare rises.
- Signing dY^*/dA positive ("more assets, more protection, more default"). The assets are kept in both options; with concave u they make repaying relatively more attractive.

Ref: Prof. Ricca, Topic 4B slides 37–44 (Aula/Topic4_B.pdf); Indarte (2023). No book exercise exists.

18. Karlan–Zinman: which comparison identifies selection and which identifies moral hazard — Original exercise

Data (full statement: list, exercise 18; built on Topic 4B slides 45–58). Loan $I = 1$; types S, R (half each); $Y(S) = 1.35$, $Y(R) = 1.80$; $\pi(S,H) = 0.9$, $\pi(S,L) = 0.8$, $\pi(R,H) = 0.7$, $\pi(R,L) = 0.5$; effort cost $c = 0.17$ for H, 0 for L (ties go to H); default cost $C_b = 0.25$ (normal future rate) or 0.45 (dynamic incentive); $r_L = 10\%$, $r_H = 30\%$; outside option 0; no strategic default. Groups: A ($r_o = r_c = r_L$), B ($r_o = r_H$, $r_c = r_L$), C ($r_o = r_c = r_H$), each split by the dynamic incentive.

What is asked. (a) Derive when a borrower works hard and apply it to both types for each contract rate and default cost. (b) Find who borrows at each offer rate, using what borrowers expect when they sign. (c) Fill in the default rate of each of the six experimental cells. (d) Pick the comparisons that change only one thing, and measure the selection and the two incentive effects. (e) Show what goes wrong when high-rate and low-rate borrowers are compared without the experiment. (f) Find the highest rate that keeps the risky type working and connect it to Holmström–Tirole.

Approach. The experiment of slides 46 and 56 turns a correlation into separate effects by randomising three things at different times. Take-up is decided at stage 1, given the *offer* rate, so the offer rate shapes the pool (hidden information). Effort is chosen at stage 2, given the *contract* rate and the dynamic incentive, so these act on behaviour (hidden action). A comparison identifies one effect if the two cells differ only in the variable that drives that effect. We solve the model backwards: effort first, then take-up, then the default table.

Notation.

- $\theta \in \{S, R\}$: type; $e \in \{H, L\}$: effort; $\pi(\theta, e)$: success (repayment) probability; $\Delta\pi_\theta \equiv \pi(\theta, H) - \pi(\theta, L)$, so $\Delta\pi_S = 0.1$, $\Delta\pi_R = 0.2$.
- r_o: offer rate (stage 1); r_c: contract rate (stage 2); C_b: borrower's cost of default, 0.25 or 0.45.
- $g_\theta(r_c, C_b) \equiv \Delta\pi_\theta[Y(\theta) - (1 + r_c) + C_b]$: gain from H before its cost c.
- d: default rate of a cell, the share of its borrowers who default, $1 - \pi$ averaged over the pool.

Step 1 — (a) The effort condition. We compare the borrower's payoff under H and under L at given r_c and C_b:

$$\begin{aligned} U(H) - U(L) &= \big[\pi(\theta,H) - \pi(\theta,L)\big]\big[Y(\theta) - (1+r_c)\big] - \big[1-\pi(\theta,H)\big]C_b + \big[1-\pi(\theta,L)\big]C_b - c \\ &= \Delta\pi_\theta\big[Y(\theta) - (1+r_c)\big] + \Delta\pi_\theta\, C_b - c \\ &= \Delta\pi_\theta\big[Y(\theta) - (1+r_c) + C_b\big] - c, \end{aligned} \tag{131}$$

so H is chosen iff $g_\theta(r_c, C_b) \geq c = 0.17$. Effort rises with the stake in success, which falls with r_c (hidden action effect 1) and rises with C_b (hidden action effect 2). The gains:

$(r_c,\ C_b)$	$(10\%,\ 0.25)$	$(10\%,\ 0.45)$	$(30\%,\ 0.25)$	$(30\%,\ 0.45)$
g_S	$0.1(0.50) = 0.05$	$0.1(0.70) = 0.07$	$0.1(0.30) = 0.03$	$0.1(0.50) = 0.05$
effort of S	L	L	L	L
g_R	$0.2(0.95) = 0.19$	$0.2(1.15) = 0.23$	$0.2(0.75) = 0.15$	$0.2(0.95) = 0.19$
effort of R	H	H	L	H

Answer (a). H iff $\Delta\pi_\theta[Y(\theta) - (1+r_c) + C_b] \geq c$. S always chooses L. R chooses H except at $r_c = 30\%$ with $C_b = 0.25$, where it chooses L.

Step 2 — (b) Take-up. At stage 1 an individual expects $r_c = r_o$ and $C_b = 0.25$ and anticipates its own effort from Step 1. The payoff $\pi[Y - (1+r_o)] - (1-\pi)(0.25) - \text{cost}(e)$ is:

$$\begin{aligned} S,\ r_o = 10\%\ (L): &\quad 0.8(1.35 - 1.10) - 0.2(0.25) = 0.20 - 0.05 = 0.15 \ \geq 0, \\ S,\ r_o = 30\%\ (L): &\quad 0.8(1.35 - 1.30) - 0.2(0.25) = 0.04 - 0.05 = -0.01 \ < 0, \\ R,\ r_o = 10\%\ (H): &\quad 0.7(1.80 - 1.10) - 0.3(0.25) - 0.17 = 0.49 - 0.075 - 0.17 = 0.245 \ \geq 0, \\ R,\ r_o = 30\%\ (L): &\quad 0.5(1.80 - 1.30) - 0.5(0.25) = 0.25 - 0.125 = 0.125 \ \geq 0. \end{aligned}$$

(The other effort gives less: R at 10% with L gets $0.5(0.70) - 0.125 = 0.225$; at 30% with H, $0.7(0.50) - 0.075 - 0.17 = 0.105$.)

Answer (b). At $r_o = 10\%$ both types borrow (take-up 100%); at $r_o = 30\%$ only R borrows (take-up 50%). The safe type leaves first: its success payoff barely covers the higher repayment, while the risky type's larger $Y(R)$ keeps the loan worth taking.

Step 3 — (c) The default table. A cell's default rate averages $1 - \pi(\theta, e)$ over the types in its pool, at the effort of Step 1 for that cell's r_c and C_b. Group A has both types; B and C only R.

Cell	Pool	No dyn. incentive ($C_b = 0.25$)	Dyn. incentive ($C_b = 0.45$)
A: $r_o = r_c = 10\%$	S, R	$\frac{1}{2}(0.2) + \frac{1}{2}(0.3) = 0.25$	$\frac{1}{2}(0.2) + \frac{1}{2}(0.3) = 0.25$
B: $r_o = 30\%$, $r_c = 10\%$	R	$1 - 0.7 = 0.30$	$1 - 0.7 = 0.30$
C: $r_o = r_c = 30\%$	R	$1 - 0.5 = 0.50$	$1 - 0.7 = 0.30$

Answer (c). Default rates: A 25% / 25%; B 30% / 30%; C 50% / 30% (without / with the dynamic incentive).

Step 4 — (d) Identification. Each comparison holds everything but one variable fixed.

- *Selection (hidden information): B vs A.* Same contract rate (10%) and same dynamic incentive; only the offer rate, hence the pool, differs:

$$d_B - d_A = 0.30 - 0.25 = +0.05 \text{ (5 pp), in both halves.} \tag{132}$$

- *Repayment burden (hidden action 1): C vs B.* Same offer rate, so the same pool (R only); only r_c differs:

$$d_C - d_B = 0.50 - 0.30 = +0.20 \text{ (20 pp) without, } 0.30 - 0.30 = 0 \text{ with the dynamic incentive.} \tag{133}$$

- *Dynamic incentive (hidden action 2): "yes" vs "no" within a group.* Same pool and same r_c; only C_b differs:

$$\text{A: } 0.25-0.25=0, \qquad \text{B: } 0.30-0.30=0, \qquad \text{C: } 0.30-0.50=-0.20\ (-20\text{ pp}). \tag{134}$$

The dynamic incentive changes behaviour only where an incentive constraint is close to binding. In C without it, R's gain 0.15 is below $c=0.17$, and $C_b=0.45$ lifts it to 0.19. In A and B, R already works ($0.19\geq 0.17$) and S is far from working (0.05 or 0.07), so a higher C_b changes nobody's choice.

Answer (d). Selection: B vs A, +5 pp. Repayment burden: C vs B, +20 pp without the dynamic incentive and 0 with it. Dynamic incentive: within C, −20 pp; 0 in A and B.

Step 5 — (e) The naive comparison. Without the dynamic incentive,

$$\begin{aligned} d_C-d_A &= 0.50-0.25=0.25 \\ &= \underbrace{(d_B-d_A)}_{\text{selection}=0.05}+\underbrace{(d_C-d_B)}_{\text{moral hazard}=0.20}. \end{aligned} \tag{135}$$

Outside the experiment the offer rate always equals the contract rate, so a high-rate borrower is both from a worse pool *and* facing a heavier repayment. There is no cell like B with $r_o\neq r_c$, so the 25 pp cannot be split. The surprise contract rate is what creates cell B.

Answer (e). 25 pp = 5 pp selection + 20 pp moral hazard; inseparable without a cell where $r_o\neq r_c$.

Step 6 — (f) The highest rate that keeps R working. From (131) with $\theta=R$:

$$\begin{aligned} 0.2\big[1.80-(1+r_c)+C_b\big] &\geq 0.17 \\ 1.80-(1+r_c)+C_b &\geq \frac{0.17}{0.2}=0.85 \\ 1+r_c &\leq 1.80+C_b-0.85. \end{aligned} \tag{136}$$

With $C_b=0.25$: $1+r_c\leq 1.20$, so $r_c\leq 20\%$. With $C_b=0.45$: $1+r_c\leq 1.40$, so $r_c\leq 40\%$. This is the Holmström–Tirole constraint $(\Delta p)R_b\geq B$ with $R_b=Y(R)-(1+r_c)$ the borrower's stake on success, c in place of B, plus the default cost: $\Delta\pi_R(R_b+C_b)\geq c$. The maximal repayment on success, $Y-c/\Delta\pi+C_b$, is the analogue of $R-B/\Delta p$, and the expected income pledgeable by a working risky borrower is $0.7(1.20)=0.84$ without and $0.7(1.40)=0.98$ with the dynamic incentive.

Answer (f). $r_c\leq 20\%$ without and $r_c\leq 40\%$ with the dynamic incentive; at 30% the IC fails only without it (cell C). The dynamic incentive raises pledgeable income by $0.7\times 0.20=0.14$.

Intuition. The offer rate works *before* the borrower is locked in, so it can only change who walks in; the contract rate and the future-loan promise work *after*, so they can only change what borrowers do. A randomised surprise between the two stages lets the economist hold the pool fixed while moving incentives, and an independent future-rate lottery separates a second incentive channel. In this example most of the naive rate–default correlation is moral hazard, and it disappears when future credit is made valuable.

Common mistakes.

- Comparing A with C and calling it adverse selection: that difference mixes the pool and the repayment burden.
- Letting borrowers anticipate the surprise: take-up uses $r_c=r_o$ and $C_b=0.25$; the lower r_c and the dynamic incentive are news at stage 2.
- Assuming the riskier type always exerts less effort. Here R works harder because it has more at stake in success ($Y(R)$ is larger); the identification does not depend on the effort ranking across types.

Ref: Prof. Ricca, Topic 4B slides 45–58 (Aula/Topic4_B.pdf); Karlan and Zinman (2009), *Econometrica* 77(6); Tirole §3.2.2, pp. 116–117 (IC$_b$, pledgeable income). No book exercise exists.

19. Reading a three-rate experiment: selection, incentives and credit rationing — Original exercise

Data (full statement: list, exercise 19; *hypothetical* numbers, not Karlan and Zinman's estimates). $r_L = 30\%$, $r_H = 40\%$ per loan. Loans and defaults: A (offered and paid 30%): 600/69 without and 600/51 with the dynamic incentive; B (offered 40%, paid 30%): 750/132 and 750/108; C (offered and paid 40%): 750/141 and 750/114. Defaulted loans repay nothing. Part (e): $I = 1$, $R = 1.5$, $p_H = 0.9$, $p_L = 0.5$, $B = 0.2$, $C_b \in \{0, 0.2\}$.

What is asked. (a) Turn the counts into default rates and read off the three effects from the right comparisons. (b) Decide which effects are statistically distinguishable from zero. (c) Map the table into the regression on slide 58. (d) Use the table to evaluate a rate increase from the lender's side and explain why it can backfire. (e) Show in the Holmström–Tirole model how a dynamic incentive relaxes the incentive constraint and loosens rationing. (f) State the limit of the design on the selection side.

Approach. The identification logic is that of slides 46 and 56: B vs A holds the contract fixed and varies the pool; C vs B holds the pool fixed and varies the repayment; the dynamic-incentive lottery is independent of everything else, so its effect can be read within each group or pooled. Because each comparison is between randomised groups, a two-sample test of proportions is valid. Part (d) applies Stiglitz–Weiss logic (Topic 3) to the measured effects, and part (e) applies the Holmström–Tirole IC (Topic 4A).

Notation.

- $\bar{d}_g$: default rate of group $g \in \{A, B, C\}$; $\bar{d}^0$, $\bar{d}^1$: pooled default rates without / with the dynamic incentive.
- $C_i \in \{0, 1\}$: dynamic-incentive dummy; $Y_i \in \{0, 100\}$: default indicator in percentage points.
- SE: standard error of a difference of proportions; z: difference over SE.
- $\mathcal{R}(r) = (1 - \bar{d})(1 + r)$: lender's expected gross repayment per unit lent.
- Part (e): R_b: borrower's pay on success; C_b: her loss on failure from forgone cheap future credit; $\Delta p = p_H - p_L = 0.4$; $\mathcal{P}$: pledgeable income; $\bar{A}$: minimum assets for financing.

Step 1 — (a) Default rates. Each rate is defaults over loans. Cells: A $69/600 = 11.5\%$ and $51/600 = 8.5\%$; B $132/750 = 17.6\%$ and $108/750 = 14.4\%$; C $141/750 = 18.8\%$ and $114/750 = 15.2\%$. Groups:

$$\bar{d}_A = \frac{69 + 51}{1200} = \frac{120}{1200} = 10.0\%, \qquad \bar{d}_B = \frac{132 + 108}{1500} = \frac{240}{1500} = 16.0\%,$$
$$\bar{d}_C = \frac{141 + 114}{1500} = \frac{255}{1500} = 17.0\%.$$

Pooled by dynamic incentive:

$$\bar{d}^0 = \frac{69 + 132 + 141}{2100} = \frac{342}{2100} = 16.2857\%, \qquad \bar{d}^1 = \frac{51 + 108 + 114}{2100} = \frac{273}{2100} = 13.0\%.$$

Step 2 — (a) The three effects. Each comparison isolates one channel.

$$\text{Selection (B vs A, same } r_c\text{):} \quad \bar{d}_B - \bar{d}_A = 16.0 - 10.0 = 6.0 \text{ pp}, \tag{137}$$
$$\text{Repayment burden (C vs B, same pool):} \quad \bar{d}_C - \bar{d}_B = 17.0 - 16.0 = 1.0 \text{ pp}, \tag{138}$$
$$\text{Dynamic incentive (yes vs no):} \quad \bar{d}^1 - \bar{d}^0 = 13.0 - 16.2857 = -3.2857 \text{ pp}. \tag{139}$$

Within groups the dynamic effect is A $8.5 - 11.5 = -3.0$, B $14.4 - 17.6 = -3.2$, C $15.2 - 18.8 = -3.6$ pp. Pooling is valid because the lottery gives exactly half of every group the incentive, so the pooled "yes" and "no" samples have the same mix of A, B and C.

Answer (a). Selection +6.0 pp (B vs A); repayment burden +1.0 pp (C vs B); dynamic incentive −3.29 pp (pooled; −3.0, −3.2, −3.6 pp within A, B, C).

Step 3 — (b) Standard errors and tests. For each difference of two independent proportions,

$$\text{SE} = \sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}.$$

$$\begin{aligned}
\text{Selection:}\quad \text{SE} &= \sqrt{\frac{0.16(0.84)}{1500} + \frac{0.10(0.90)}{1200}} = \sqrt{0.0000896 + 0.0000750} \\
&= \sqrt{0.0001646} = 0.012830, \\
z &= \frac{0.06}{0.012830} = 4.677, \qquad \text{CI} = 0.06 \pm 1.96(0.012830) = [0.034854,\ 0.085146]. \\
\text{Burden:}\quad \text{SE} &= \sqrt{\frac{0.17(0.83)}{1500} + \frac{0.16(0.84)}{1500}} = \sqrt{0.0000941 + 0.0000896} \\
&= \sqrt{0.0001837} = 0.013552, \\
z &= \frac{0.01}{0.013552} = 0.738, \qquad \text{CI} = [-0.016563,\ 0.036563]. \\
\text{Dynamic:}\quad \text{SE} &= \sqrt{\frac{0.13(0.87)}{2100} + \frac{0.162857(0.837143)}{2100}} = \sqrt{0.0000539 + 0.0000649} \\
&= \sqrt{0.0001188} = 0.010899, \\
z &= \frac{-0.032857}{0.010899} = -3.015, \qquad \text{CI} = [-0.054218,\ -0.011496].
\end{aligned}$$

Answer (b). Selection ($z = 4.68$, CI 3.5 to 8.5 pp) and dynamic incentive ($z = -3.01$, CI −5.4 to −1.1 pp) are significant at 5% ($|z| > 1.96$); the repayment burden ($z = 0.74$, CI −1.7 to 3.7 pp) is not.

Step 4 — (c) The slide-58 regression. The design has three rate groups, and the constant, r_o and r_c give exactly three free parameters, so OLS fits the three group means exactly. C_i is balanced (one half in every group), so it is orthogonal to the group structure. By the Frisch–Waugh–Lovell theorem, β_b is the regression of Y on $C_i - \frac{1}{2}$, which is the n-weighted average of the within-group differences:

$$\beta_b = \frac{1200(-3.0) + 1500(-3.2) + 1500(-3.6)}{4200} = \frac{-3600 - 4800 - 5400}{4200} = \frac{-13800}{4200} = -3.285714,$$

the pooled difference (139). Each group's fitted value, averaged over its two halves, equals its mean: $\alpha + \beta_o r_o + \beta_c r_c + \frac{1}{2}\beta_b = \bar{d}_g$. Subtracting the equations of A and B (they differ only in r_o) and of B and C (only in r_c):

$$\begin{aligned}
\beta_o(40-30) = \bar{d}_B - \bar{d}_A = 6.0 &\implies \beta_o = 0.6, \\
\beta_c(40-30) = \bar{d}_C - \bar{d}_B = 1.0 &\implies \beta_c = 0.1, \\
\alpha = \bar{d}_A - \tfrac{1}{2}\beta_b - 30(\beta_o + \beta_c) &= 10 + 1.642857 - 21 = -9.357143.
\end{aligned}$$

(Solving the 4×4 normal equations directly gives the same four numbers; the negative intercept is an extrapolation to zero rates, outside the data.)

Answer (c). $\beta_o = 0.6$ pp of default per pp of offer rate (hidden information, adverse selection); $\beta_c = 0.1$ (hidden action 1, not significant); $\beta_b = -3.29$ pp (hidden action 2); $\alpha = -9.36$.

Step 5 — (d) The lender's return. With defaulted loans repaying nothing,

$$\mathcal{R}(30\%) = (1 - 0.10)(1.30) = 1.170, \qquad \mathcal{R}(40\%) = (1 - 0.17)(1.40) = 1.162. \tag{140}$$

Decompose the change by adding and subtracting $(1-\bar{d}_A)(1+r_H)$:

$$\begin{aligned}\mathcal{R}(40\%)-\mathcal{R}(30\%) &= (1-\bar{d}_A)(r_H-r_L)-(\bar{d}_C-\bar{d}_A)(1+r_H)\\ &= \underbrace{0.90(0.10)}_{\text{rate}=+0.090}-\underbrace{0.06(1.40)}_{\text{selection}=0.084}-\underbrace{0.01(1.40)}_{\text{burden}=0.014}=-0.008.\end{aligned}$$

Had the pool and behaviour stayed as in A, the higher rate would have yielded $0.90(1.40)=1.26$.

Answer (d). 1.170 at 30% vs 1.162 at 40%: the rate increase lowers the expected return by 0.008 (+0.090 rate, −0.084 selection, −0.014 burden). The return is non-monotonic in the rate, so the loan supply curve bends backward: a lender facing excess demand at 30% does better to ration than to raise the rate.

Step 6 — (e) Holmström–Tirole with a dynamic incentive. (i) The borrower gets R_b on success and loses C_b on failure. Behaving beats misbehaving iff

$$\begin{aligned}p_H R_b-(1-p_H)C_b &\geq p_L R_b-(1-p_L)C_b+B\\ (p_H-p_L)R_b+[(1-p_L)-(1-p_H)]C_b &\geq B\\ (\Delta p)(R_b+C_b) &\geq B. \qquad (141)\end{aligned}$$

(ii) The minimum stake is $R_b=B/\Delta p-C_b$ (non-negative for these values), so lenders can be promised $R-B/\Delta p+C_b$ on success:

$$\mathcal{P}(C_b)=p_H\Big(R-\frac{B}{\Delta p}+C_b\Big),\qquad \frac{B}{\Delta p}=\frac{0.2}{0.4}=0.5.$$

For $C_b=0$: $\mathcal{P}=0.9(1.5-0.5)=0.9$ and $\bar{A}=I-\mathcal{P}=1-0.9=0.1$. For $C_b=0.2$: $R_b\geq 0.5-0.2=0.3$, $\mathcal{P}=0.9(1.5-0.5+0.2)=0.9(1.2)=1.08$, and $\bar{A}=1-1.08=-0.08<0$: every borrower, even one with $A=0$, is financed. Behaving is worth inducing: the NPV is $p_H R-I=1.35-1=0.35>0$, while misbehaving gives $p_L R+B-I=0.75+0.2-1=-0.05<0$. (iii) With competitive lenders the borrower gets the NPV less the expected deadweight loss:

$$U_b=0.35-(1-p_H)C_b=0.35-0.1(0.2)=0.35-0.02=0.33.$$

(iv) The repayment-burden effect is the IC (141) at work: a higher contract rate raises the lenders' claim and lowers R_b, and borrowers whose IC was nearly binding switch to misbehaving. That is why lenders cannot raise repayments beyond $R-B/\Delta p$ and why rationing starts below $\bar{A}$. Its small, insignificant size here says that at these rates few borrowers sit at the margin. The dynamic-incentive effect is C_b in (141): future cheap credit is a punishment for default that costs the lender nothing today, raises pledgeable income, and lowers $\bar{A}$. The significant −3.3 pp says this channel is strong, and that bankruptcy rules or renegotiation that soften the punishment (slides 37–40) cut pledgeable income.

Answer (e). IC: $(\Delta p)(R_b+C_b)\geq B$. $\mathcal{P}=0.9$, $\bar{A}=0.1$ without; $\mathcal{P}=1.08$, $\bar{A}=-0.08$ (no rationing) with $C_b=0.2$; borrower's utility 0.33.

Step 7 — (f) What the design cannot say. The selection effect compares pools formed at different offer rates, and people differ in two hidden ways when they select: in risk type and in the effort they *plan* to exert. Both are hidden information at take-up, and the design cannot separate them (slide 47). A borrower who accepts 40% because she already plans to put in little effort produces the same +6 pp as a borrower of a riskier type.

Answer (f). It cannot separate selection on unobserved risk from selection on unobserved anticipated effort, so a significant β_o shows hidden information, not necessarily adverse selection on risk type.

Intuition. Three randomisations give three clean contrasts. In these invented data the pool deteriorates sharply when the offer rate rises, which is the lemons and Stiglitz–Weiss mechanism

of Topic 3: price cannot clear the market because it changes the quality of the borrowers. The incentive side shows that the repayment burden matters little at these rates, but future credit is a powerful carrot, the Holmström–Tirole IC in action. A lender that learns this should keep the rate low, ration, and lean on dynamic incentives.

Common mistakes.

- Testing the selection effect with C vs A: that difference (7 pp) also contains the repayment burden.
- Concluding "no moral hazard" from the insignificant burden effect: the dynamic-incentive effect is hidden action too, and it is significant.
- Reading $\mathcal{R}$ as profit: it is gross repayment per unit lent; what matters for rationing is that it *falls* with the rate.

Ref: Prof. Ricca, Topic 4B slides 45–58 (Aula/Topic4_B.pdf); Topic 3 slides 4–19 and 47–55 (lemons, Stiglitz–Weiss; Aula/Topic_3.pdf); Tirole §3.2.2, pp. 116–117. Data hypothetical. No book exercise exists.